

KENYATTA UNIVERSITY

DIGITAL SCHOOL OF VIRTUAL AND OPEN LEARNING

IN COLLABORATION WITH

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MEDIA PRACTICALS AND PEER TEACHING

WRITTEN BY: Dr. David O. Oludhe
Mr. Lordvicius W. Olanga
Dr. Florence Miima
Ms. Mary N. Gikonyo

VETTED BY: **PEBL Quality Assurance Team Members**
Dr. George Onyango
Dr. Samuel Mutweleli
Dr. Rhoda Gitonga
Dr. Elizabeth Mwaniki
Dr. Eric Masika
Dr. Hellen Kiende

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INTRODUCTION

Welcome to this module for the unit, **Media Practicals and Peer Teaching**. This unit is special in three respects. First, it is a practical unit in that you will be required to practise various skills and also your assessment will for the most part be practical. Second, this unit is central to your training as a teachers as it involves the practical application of what you will have covered in your education and subject area units. Third, the unit is made up of two fairly distinct areas; namely, **Media practicals** and **Peer teaching**. In Media practicals, you will be involved in designing, developing, and using various teaching/learning resources; and in Peer teaching, you will be preparing micro lessons on a variety of teaching skills with the view to practicing the skills in a simulated environment.

It is hoped that you will find this unit not only interesting but also crucial to your professional development. Welcome.

MEDIA PRACTICALS AND PEER TEACHING: FLOW CHART

WEEK	TOPIC
WEEK 0:	INTRODUCTION (YOUR CONTEXT, YOUR GOALS, THOUGHTS ABOUT TEACHING AND THE USE OF TEACHING LEARNING RESOURCES)
WEEK 1:	INTRODUCTION TO MEDIA PRACTICALS AND PEER TEACHING
WEEK 2:	LECTURE SKILL
WEEK 3:	QUESTIONING AND REINFORCEMENT
WEEK 4:	SET INDUCTION AND STIMULUS VARIATION
WEEK 5:	SMALL GROUP AND USE OF EXAMPLES
WEEK 6:	INTEGRATED SKILLS
WEEK 7:	WRITING AND INTERACTIVE BOARDS
WEEK 8:	INSTRUCTIONAL GRAPHIC MATERIALS
WEEK 9:	AUDIO AND AUDIO-VISUAL PRODUCTION
WEEK 10:	COMMUNITY RESOURCES WITH EMPHASIS ON REALIA
WEEK 11:	CONSTRUCTION SKILLS
WEEK 12:	MANAGEMENT OF LEARNING RESOURCE CENTRES (LRC's)

OVERVIEW OF THE COURSE

Week 0: Introduction (Your Context, Your Goals)

This lesson is intended to help you acclimatize to blended learning and to create a community of learners who will motivate each other during the course. You will be required to introduce yourself to your lecturer and colleagues either physically during a face to face session or even online before other academic interactions start. This will be at the discretion of individual universities and lecturers. It will be important to also state your context and goals as well as what you think about businesses. You can also share any experience that you may have regarding businesses.

Week 1: Introduction to Media Practicals and Peer Teaching

In this first lesson, we explain in detail the nature of this unit; more specifically, we seek to enable you understand the specific requirements for each of the two fairly distinct areas (Media Practicals and Peer Teaching) covered in this module.

Week 2: Lecture Skill

In almost all lessons, a teacher has to present information and ideas at various points in a lesson. S/he introduces topics, summarizes important points, and stimulates further learning, and so on. All these activities require the use of lecture/explanation skill. This lesson, instruct you on how to use this skill effectively.

Week 3: Questioning and Reinforcement

Teachers rely heavily on question-asking a major part of their teaching repertoire. In this lesson, we attempt to show you how to use the questioning skill in a purposive and controlled manner to bring about effective learning. Related to question asking is how a teacher reacts to responses received from the learner. Here, we explain the skill of reinforcement for purposes of increased learning.

Week 4: Set Induction and Stimulus Variation

Effective learning requires that the interest of the learner is not only captured but also sustained. The two skills of set induction and stimulus variation respectively attend to how to stimulate/capture interest at the start of a lesson or in an activity during a lesson and how to sustain the interest in the course of a lesson.

Week 5: Small Group-work and Use of Examples

Research invariably show that learning is improved when learners learn among themselves in small groups in a planned manner. In this lesson, you will learn about the various small group formations and circumstances in which they lend themselves to their use in class. The lesson also instructs on illustrating lessons with suitable examples in form of concepts, objects, and specimens and so on.

Week 6: Integrated Skills

This lesson is a culmination of all the peer teaching skills you have been learning in previous lessons. You can now use all the peer teaching skill you have learnt. This lesson also deals with the decision a teacher requires to make in selecting and using the different skills at his/her disposal.

Week 7: Writing and Interactive Boards

In this module, lesson seven (7) all through to lesson twelve (12) involve Media practicals. This lesson deals with procedures and practices of using various writing boards in the teaching/learning process.

Week 8: Instructional Graphic Materials

Graphic instructional materials are among the most useful and versatile teaching aids at the disposal of a teacher. In this lesson you are informed of the various types of graphic materials and the circumstances in which they lend themselves. More importantly, this lesson helps you in the design, production and use of graphic materials.

Week 9: Audio and Audio-visual Production

This lesson outlines the procedures of designing, producing and using audio and audio-visual media.

Week 10: Community Resources with emphasis on Realia

Real life experiences offers opportunity for effective learning. One way of engaging learners in real life experiences is through the use of community resources. This lesson focuses specifically with one component of community resources; this being, realia (real things, objects or specimens). It illustrates how to systematically plan for, collect, research, prepare and present realia.

Week 11 Construction Skills

When it is not possible to use realia (real things, objects and specimens) to illustrate your lessons for whatever reason/s, then your next best option is to design, produce and use a model. In this lesson, tips are given on materials and production techniques one can use in preparing models for instruction in class.

Week 12 Management of learning Resource Centre (LRC's)

As resources are requested for, prepared, shared, collected and accumulated, it is essential to provide storage and other services connected to teacher support retrieval of the materials in schools. This necessitates the setting up of LRC's. Various aspects of LRC's management issues are discussed in this lesson.

Troubleshooting

○ If you have problems accessing or logging into the system (LMS):

1. Go to the Student Help Desk -<https://support.ku.ac.ke/>
2. Chat with us at <http://chat.ku.ac.ke/>
3. Call the Digital school offices at (See our Contacts here) - <http://www.ku.ac.ke/dsvol/contact-us>
4. Guides about how to use moodle are available at <http://www.ku.ac.ke/dsvol/students-resources>
5. Also use our social media huddles:

Twitter: @KUDigitalSchool

Facebook: @KUDSVOL

MODULE LEARNING OUTCOMES

By the end of this module, you will be able to:

1. Apply communication process in the teaching/learning process.
2. Employ appropriate verbal/non-verbal communication in teaching.
3. Guide learners in developing visual literacy skills.
4. Develop and use Visual Literacy materials.
5. Develop and use digital content to promote learning.
6. Use various resources and modes in teaching learners in distance education.
7. Use multichannel approach in the teaching/learning process.
8. Apply the concept of mediated learning in the teaching/learning process.
9. Integrate community resources in the teaching/learning process.
10. Effectively plan and use instructional media.
11. Manage a learning resource centre.
12. Collaborate with mentor in designing instruction and developing teaching strategies.

COURSE DESCRIPTION

This unit meant for all students taking Bachelor of Education degree and postgraduate Diploma in Education courses. It is a wholly practical unit where much of theory learnt in educational units in all education departments finds practical application. It involves practicing a variety of skills under two fairly distinct components; namely, Media Practicals and Peer Teaching. Under Media Practicals, skills involving design, production and use of instructional resources are developed; and under Peer Teaching, a variety of specific teaching skill are progressively learnt and then each followed by actual teaching of peers in using micro lesson plans prepared specifically for each skill. This unit precedes teaching practice (TP) in schools and learning institutions countrywide.

COURSE REQUIREMENTS

This is a blended learning course that will utilize the flex model. This means that learning materials and instructions will be given online and the lessons will be self-guided with the lecturer being available briefly for face to face sessions and support and also on-site (online) most of the time. Your lecturer will be meeting you face to face to introduce a lesson and put it into perspective and you will actively participate by undertaking several online activities. This means that some of the 39 instructional hours of the course will be delivered face to face while other lessons will be taught online through various learner and lecturer activities. It is important for you to note that one instructional hour is equivalent to two online hours. Three instructional hours will be needed per week. Out of these, one will be used for face to face contact with your lecturer (also referred as e-moderator in the online activities) while the other two instructional hours (translating to four online hours) will be used for online activities otherwise referred to as e-tivities in the lessons. This will add up to the 5 hours requirement per lesson earlier mentioned. You are advised to follow the topic flow-chart given so that you cover at least a lesson every week.

You will be required to participate and interact online with your peers and the e-moderator who in this case is your lecturer. Guidelines for the online activities (which we shall keep referring to as e-tivities) will be provided whenever there is an e-tivity. Please note that since the online e-tivities are part of the learning process, they may be graded at the discretion of your e-moderator. Such grading will however be communicated in the e-tivity guidelines and feedback given as soon as possible after the e-tivity. The e-tivities will include but will not be limited to online assignments and discussions. There are also assessment questions that you can attempt at the end of every lesson to test your understanding of the lesson. The answers to all the assessment questions are at the end of the module after lesson 12.

ASSESSMENT

It is important to note that the module has embedded certain learner formative assessment feedback tools that will enable you gauge your own learning progress. The tools include online collaborative discussions forums that focus on team learning and personal mastery and will therefore provide you with peer feedback, lecturer assessment and self- reflection e-tivities will account for 30% of your final examination score with the remaining 70% coming from a face to face assessments in your practical projects involving productions of instructional resource materials and peer teaching lesson presentations.

We wish you the very best of experiences in this course.

TABLE OF CONTENTS

Lesson	Page
1. Introduction to the Unit, Media Practicals & Peer Teaching	
Mr. Lordvicus Olanga-----	1
2. The Lecture Skill	
Ms. Mary Gikonyo-----	7
3. Questioning & Reinforcement Skills	
Dr. Florence Miima-----	10
4. Set Induction & Stimulus Variation Skills	
Ms. Mary Gikonyo-----	20
5. Questioning & Reinforcement Skills	
Dr. Florence Miima-----	25
6. Set Induction & Stimulus Variation Skills	
Ms. Mary Gikonyo-----	32
7. Small Group-work & Use of Examples Skills	
Dr. Florence Miima-----	35
8. Integrated Skills	
Ms. Mary Gikonyo-----	30
9. Writing & Interactive Boards Skills	
Dr. Florence Miima-----	34
10. Graphics Skills	
Dr. David Oludhe-----	40
11. Audio & Audio Production Skills	
Mr. Lordvicus Olanga-----	47
12. Realia & Community Resources	
Dr. David Oludhe-----	66
13. 3-D Construction Skills	
Dr. David Oludhe-----	69
14. Organizing & Managing Learning Resource Centres (LRC's)	
Mr. Lordvicus Olanga-----	74
15. Answers to Lesson Assessment Questions	89

KENYATTA UNIVERSITY

Media Practical & Peer Teaching

LESSON ONE

INTRODUCTION TO THE UNIT

1.1 Introduction

This module (ECT 300/ECT703: Media practical and peer teaching) assumes that the earlier units namely ECT 201/ECT701: Contemporary pedagogies in Education, ECT 202/ECT702: Introduction to Instructional design and Technology and ECT 203: Instructional research and practice are prerequisites. This unit will be a prerequisite for the subject methods units, ECT 331: Teaching Practicum 1 and ECT 301: Teaching Practicum 2.

1.2 Lesson Learning Outcomes

By the end of this lesson, the teacher trainee will be able to:

1. Employ appropriate verbal / nonverbal communication in teaching
2. Develop and use visual literacy materials
3. Guide learners in developing visual literacy skills
4. Develop and use digital content to promote learning
5. Use various resources and modes in teaching learners in distance education
6. Use multi-channel approach in the teaching/ learning process
7. Apply the concept of mediated learning in the teaching / learning process
8. Integrate community resources in the teaching / learning process
9. Effectively plan and use instructional media
10. Manage a learning Resource Centre
11. Collaborate with mentors in designing instruction and developing teaching strategies

1.2.1 Media practical

Before you start a lesson, you plan the lesson, which is a process that involves using an instructional design or model. Effective teaching with technology also involves using an instructional model. An instructional model is a systematic guide for planning instruction or a lesson. When using technology, an instructional model and planning takes on a more important role. The technology the teacher decides to use is not the critical issue, what is important is that s/he uses the technology effectively and the technology is appropriate to the expected learning outcome. Many instructional models are available from which you can choose. One such model, the ASSURE model is covered in ECT 202/ECT702.

The first part of the module shall look at media practical. Teaching and generally instruction should be supported by the use of resources. There is a wide variety of teaching / learning resources. The first part of this module shall consider the preparation and/or acquisition of some of these resources. The use of resources is set on the premises that learning requires the use of most and if possible, all the five senses. Different learners have individual learning preferences and therefore would benefit from a multi-sensory approach to resource planning and utilization. The teacher should develop innovative ways to reach a diverse population of learners with different learning styles. A learning style refers to how individuals learn, including how they prefer to receive information, process the information and retain the information. In the long run, teachers want their learners to be able to assimilate information and become problem solvers. In this module we shall consider the most commonly utilized resources. These include:

1. Chalkboard and PowerPoint presentations
2. Preparation of graphic materials
3. Preparation of sound recordings
4. Acquisition and use of 3 dimensional materials
5. Utilizing and managing learning resources

1.2.2 Introduction to Micro Teaching

The second part of the module shall look at micro (peer teaching).

Microteaching is a teaching training development tool that allows the trainer to review the teaching techniques of a teacher trainee in an experimental set up during the training. The trainer gives the trainee an opportunity to teach short lessons for short periods of time as s/he appraises and /or critiques the performance. Microteaching simplifies the complexity of teaching by breaking it down into several skills to be practiced in isolation with time, class size and task scaled down to provide an optimal training environment. The appraisal / critiquing gives the trainee confidence to face a larger class for a longer period of time in future besides realizing his/her mistakes and improving on them.

In this module, we are going to use the terms microteaching and peer teaching interchangeably because besides being small scale teaching, the trainees are also teaching fellow undergraduates who are their peers. Again, unlike in the normal classroom, the trainees are also partly appraised and/or critiqued by their peers who form the class.

1.2.3 Brief History of Micro Micro-Teaching

Micro-teaching (MT) was started by Professor Dwight Allen at Stanford University in 1963 as a way of exposing teacher-trainees to broken-down teaching skills one at a time. It has been used to train teachers at all levels of education by breaking down teaching into practicable skills. Over the years with massive enrolments and poor facilities it has been rendered less efficient in the developing world. The original MT model required TEACH – VIEW- CRITIQUE – RETEACH - OBSERVE –CRITIQUE cycle for each student teacher for each skill. In other words, the steps involved were

1. Trainer gives background information and demonstrates the skill to be practiced
2. The class is then shown a demonstration video of the skill
3. A trainee picks a sub topic from one of his/her subjects that can be used to demonstrate the skill under discussion. S/he then prepares a micro lesson of between 7 to 10 minutes in the usual lesson plan format namely administrative details, specific learning outcomes, time allocation over the three stages (namely introduction, main body and conclusion), and content, learning activities, resource materials both print and non-print. The trainee should not attempt to squeeze the content of 40 minutes into the 7-10 minutes but should target a small section of the content.
4. The micro lesson is presented by the trainee with the trainer and peers keenly observing and noting strong points and weak points for discussion. The presentation where possible should be recorded for playback and immediate feedback.
5. Once the presentation is over, the video recording should be played back for the trainee to also observe his/her presentation with the view of appraising and/or critiquing it
6. The critiquing should be done by the trainee, peers and trainer in that order
7. The trainee should implement the comments and prepare to re-teach the same content maybe after a week for comparison
8. This time round the critiquing should be targeting improvements on points raised during the first presentation

An individual trainee should go through steps 1-8 for each of the skills. The skills are as follows:

1. Lecture / explanation skill
2. Questioning skill
3. Reinforcement skill
4. Set induction skill
5. Stimulus variation skill
6. Small group work skill
7. Use of examples / illustrations
8. Integration of the skills

1.2.4 General Expected Outcomes of Micro (Peer)-Teaching

Micro (peer)-teaching aims at achieving the following learning outcomes:

1. Preparation of student-teachers for the actual teaching practice
2. Reduction of the complexity of the teaching task by allowing trainee-teachers to concentrate on the practice of specific lesson preparation and presentation skills
3. Provision of a means of easing the tension on the trainee-teachers with less trauma, from the theory of methods to the realism of the practical classroom situation
4. Development of critical observation of what constitutes effective teaching.

1.2.5 Importance of micro teaching

Peer teaching is important in teacher training because:

1. It provides practice of teaching skills without the pressure of actual teaching situation
2. It provides opportunity for immediate feedback from the trainer and the peers
3. It enables the teacher trainees to improve their teaching skills
4. The teacher trainee gains confidence in teaching and in facilitating learning
5. It prepares one for teaching practice / internship
6. It involves teaching one's peers thereby improving collaboration
7. The performance of the teacher trainee is noted down by the peers and the trainer and then reviewed or discussed. The peers also learn in the process.
8. Once the various skills have been practiced and mastered in isolation, they are eventually integrated and practiced for a slightly longer time (about 15 -20) minutes.

Micro-teaching provides the bridge between educational theories and classroom teaching in 3 phases. The 3 phases are:

a. Acquisition of knowledge.

Each skill is discussed and analyzed in a general lecture after which a film/video is shown. In the film / video an experienced teacher demonstrates the use of the skill. **b. Acquisition of skills.**

Each student –teacher after careful study of the skill prepares a micro-lesson of duration 7- 10 minutes, a carbon copy of which should be made for the supervisor at the beginning of each practice session. The members of the peer group play the role of the pupils. As much as possible each learner should get an opportunity to prepare and present each of the skills. It is recommended that if time and facilities allow for the tutor to arrange for a video recording, playback and re-teach session so as to address problem areas. **c. Transfer of skills.**

After all the skills have been practiced separately, an integrated skill practice of approximately 15 to 20 minutes is introduced which aims at integrating the various skills. The supervisor evaluates the performance of the student-teacher and gives feedback.

1.2.6 Challenges Faced by Training Institutions

Teacher training institutions face several challenges in utilizing micro teaching as a training tool. These challenges include:

1. Increased enrolment in teacher training institutions outstripping facilities
2. Inability to get trainees to be in small groups of 5 and to go through the training cycle of

TEACH – VIEW- CRITIQUE – RETEACH-OBSERVE –CRITIQUE for each of the skills due to lack of time and facilities. It is usual to find micro teaching groups as large as 100 trainees thus rendering it impossible to apply the original Dwight Allen model.

3. Lack of recording facilities to enable proper use of the micro-teaching model.

1.2.7 The Lesson Plan/Outline Format

The micro-lesson plan has similar features to the actual lesson plan for the full lesson. It is made up of:

- The normal administrative details
- Specific instructional learning outcomes clearly stated and appropriate for the class level. Note that in a normal lesson, the learning outcome determine the teaching approach while in micro-teaching the skill determines the learning

outcomes. As much as possible the learning outcomes should span the various domains and the levels therein.

- Introductory step: clearly showing a guide to the skill to be demonstrated
- Lesson development step: Clear and accurate information on the content to be covered
- Concluding step: Clear statement concerning the manner in which the conclusion of the lesson is to be brought about bearing in mind the content

Horizontally, the lesson plan format caters for time allocation, content in note form, learning activities in terms of the learner and in proper variety, and resources. The references should be various and clearly show the title, author and page.

1.3 Assignment

Prepare a 10-minute micro lesson for the skill of lecture/explanation. Submit the lesson plan to your lecturer for assessment and feedback.

1.4 E-References

1. https://www.youtube.com/results?search_query=how+to+make+an+online+class+interesting
2. <https://www.youtube.com/watch?v=hg4LN9hKdH0&t=80s>
3. <https://www.youtube.com/watch?v=hFNkBsJca0c&t=299s>
4. <https://www.youtube.com/watch?v=hFNkBsJca0c&t=302s>

LESSON 2 THE LECTURE SKILL

2.1 Introduction

The lecture skill is an organized presentation of subject matter – usually with assistance of visual aids.

It is an uninterrupted talk from the teacher to the students. In other words, one-way communication from the active presenter to a usually passive audience. However, There are two forms of lecturing namely, formal and informal. A formal lecture is purely verbal and the communication is basically one-way from the teacher to the students without the students interrupting the exposition while informal lecture allows students to ask/answer questions, make comments, suggestions and the teacher to demonstrate, illustrate, etc. In schools, informal lecture is ideal because it allows constant dialogue between the teacher and the students besides using learning resources or teaching aids.

2.2 Lesson Learning Outcomes

By the end of the lesson, you should be able to:

- 1.2.1 Define the concept of lecturing skill.
- 1.2.2 Explain the two types of lecturing skill
- 1.2.3 Demonstrate how you would use lecturing skill in the classroom.

2.2.1 The Concept of Lecturing

Lecturing is a skill that calls for verbal and non-verbal communication from the teacher to the students or from a speaker to the audience. General terms, Lecture method is a process of verbally transmitting a body of knowledge to a recipient in accordance to a pre-planned scheme.

2.2.2 Purpose of Using Lecture Skill in Teaching

In teaching and learning process the teacher may justifiably use the lecture skill when:

- 1. Introducing a new topic or unit.
- 2. Presenting materials not easily obtainable.
- 3. Supplementing textbook materials.
- 4. Informing students of the expected results of a learning activity.
- 5. Attempting to cover a great deal of materials within a short time.

6. Summarizing important points of a lesson of study

E-tivity 2.2.1 Meaning of Lecture Skill

Numbering, pacing & Sequencing	2.2.1
Title	Meaning of Lecture Skill
Purpose	The Purpose of this activity is to define in your words the term Lecture Skill.
Brief summary of overall task	Go to https://www.researchgate.net . Open [Pdf] Effective Teaching Methods. Go to www.google CTE – Center for Teaching Excellence to learn more on Lecturing Methods. Click prep..">https://poorvucenter.yale.edu>prep..
Spark	
Individual Contribution	a) Define Lecturing Skill in teaching and learning situation. b) Explain the two types of Lecturing. c) State the reason which would make you use either of the skill stated in the classroom.
Interaction begins	d) Explain and post the circumstances that would justify your use of lecturing skill in the classroom. e) State two advantages and two disadvantages of using lecturing in the classroom.

E-moderator interventions	• Focus on learner's concentration on the content. • Encourage further reading on the topic from the internet. • Give feedback as at intervals. • Invite questions for clarity if necessary. □ Close the E-tivity
Schedule and time	The task should take 30 minutes
Next	Advantages and Disadvantages of Lecture

2.3 Assignment

1. Prepare a 40 minutes lesson plan on the Lecture Skill.
2. State how you can improve on the delivery of a lecture for better learning.

2.4 References:

Laughton, Dave (2011) – What is the use of Lectures?

Pdf File – What is the use of lectures – A comprehensive guide to the uses and abuses of the lecture method.

World Bank (1994) – Higher Education – The lessons of experience. Washington DC.

Gibbs, G. & T. Habeshaw (1989) – Preparing to Teach – An introduction to effective Teaching in Higher Education. Bristol-Technical & Educational Services.

Goffe & Kauper (2014) – Knowing how to lecture well is crucial.

Murray in Perry & Smart (1997) – Effective lecturing is characterized by enthusiasm.

Wammes, J.D. et al (2016) – Mind wandering during lectures. Changes in rates across an entire semester. Scholarship of Teaching and Learning in Psychology, 2(1), 13-322.

LESSON 3

QUESTIONING AND REINFORCEMENT SKILLS

3.1 Introduction

In this lesson, we will define the term questioning and reinforcement and discuss how the two concepts can be used in teaching and learning process. Questioning and reinforcement skill is key in the classroom instructional practices and anyone can learn to use them well. It's a way of helping teachers and learners develop their own ability to raise and formulate questions. In Teaching and learning process, teachers rely heavily on question asking as a major part of their teaching repertoire. It is one of the most versatile and most readily available technique to the teacher. It must be an action that is clearly observable, recognizable and purposive. The teacher must perform it in a controlled way. In this skill the teacher controls all the learning activities, knowing from the beginning where to lead the learners through the questions. The purpose of this lesson is to enable you understand the concepts and how you can apply them in the teaching and learning process. This lesson is in two parts: a) questioning and b) reinforcement

3.2 Lesson Learning Outcomes

By the end of this lesson you will be able to:

3.2.1 Explain the concepts of questioning and reinforcement skills.

3.2.2 Use different types of questions and reinforcement in the classroom instruction.

3.2.3 Demonstrate how to use questioning technique in the teaching and learning process.

3.2.4 Use reinforcement in classroom instruction.

3.2.1 The Concept of Questioning

Questioning is a skill that involves asking questions to elicit information or test a person's knowledge or ability. Questions require responses that are well thought and therefore in the teaching and learning process it enhances learner's participation in the classroom.

The purposes of using Questioning skill in the Teaching/learning

Process Teachers use questions for several reasons and some of them are:

1. Clarification e.g. Ask the learners to clarify some concepts or information and also to check on prior knowledge and understanding.
2. Enhancing inter action between the teacher and the learners. This helps in creating interest, engaging and challenging the learners.
3. Increasing learner's critical awareness hence help them to extend their thinking from the concrete and factual to the analytical and evaluative
4. Refocusing /focus learner's' thinking on key concepts, related issues and lesson outcomes
5. Prompting/ e.g. The teacher gives hints or clues to the correct response
6. Redirecting e.g. Ask the same questions to other learners where the response is incorrect, incomplete or wrong
7. Evaluating the learner's level of understanding for feedback.

E-tivity 3.2.1 Concept of Questioning

Numbering, pacing and sequencing	3.2.1
Title	Definition of the term questioning
Purpose	The purpose of this e-tivity is to enable you to explain the meaning of questioning
Brief summary of overall task	Click here to read the material and come up with the clear definition of questioning. b) Summarize the reasons why the teachers use questioning skill in the classroom instruction.
Spark	
Individual task	(a) Point out the key terms in the definition of questioning (b) State any four reasons why questioning skill is important in the classroom instruction.
Interaction begins	a) Post three key words that describe questioning b) State the purpose of questioning in teaching and learning process.

E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. □ Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	Types of questions

3.2.2 Types of questions

1. Lower order questions (knowledge questions). The learner is required to recall facts, give definitions, make observations etc. The questions are derived from existing knowledge or from prior knowledge. The learners are not expected to use their ideas, modify or develop new knowledge. Some of the questions may require “yes” and “no” responses
2. Middle order questions. (comprehension and application questions) The questions require the learners to apply known principals, rules or techniques to solve problems. They should be able to compare, contrast, translate apply and distinguish. The questions are more demanding than the lower order questions and imply that learners respond by using existing knowledge to apply to new context.
3. Higher order questions (Analysis, synthesis and evaluation). The questions require the learners to go beyond what they know. The questions are intended to engage the learners in the skill of thinking critically and creating knowledge. The learners may be involved in solving problems, making judgements, making reflections, making predictions, justifying issues etc. They need intelligent manipulation of learning materials.

E-tivity 3.2.2 Types of questions

Numbering, pacing and sequencing	3.2.2
Title	Types of questions
Purpose	The purpose of this e-tivity is to enable you to identify types of questions used in the classroom instruction
Brief summary of overall task	Watch this video link and identify any four types of questions.

Spark	
Individual task	(a) Using bullet points, outline the key terms/verbs used to identify different types of questions. (b) Formulate a question for each of the three types of questions identified in one of your subject areas.
Interaction begins	c) Post the three types of questions that are commonly used by teachers in the classroom d) Explain why it is advisable for a teacher to use different types of questions in the classroom instruction.
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. □ Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	When and how to ask questions

3.2.3 When and How to Ask Questions.

Questions are used throughout the lesson right from when the lesson begins to the end. At the beginning of a lesson, the questions are used to determine the starting point, create atmosphere of wonder, curiosity and interest. They can also be used as set induction. During presentation of a lesson, questions can be used to stimulate the learners thought, make them active or to enable them participate in lesson development. They can also be used to retain the interest of the learners and even discover their misconceptions. At the end of a lesson, questions are used to summarize the lesson, emphasize the main point and the purpose of a lesson and also to allow the learners draw their own conclusions.

Characteristics of Good Questions

The questions are initiated by the teacher and require the learner to think before responding and therefore they should be:

- 1 Relevant- focus on recall of the content taught and aligns well with the learning outcome

- 2 Clear- should be framed in a clear, easily understandable language.
- 3
- 4 Concise-the learners should understand what is required from the question. A good question should avoid any unnecessary information that may make the learners spend time understanding it.
- 3 Purposive-focus on the learning outcome.
- 4 Guiding but not leading; guide the learners to understand the concept.
- 5 Stimulate thinking- Require the learners to think through and recall the concepts taught.
- 6 Single dimension-focus on one dimension at a time.
- 7 Suited to the age, interest and ability of the learners.

Procedure of asking questions

A teacher should Plan to use questions that encourage thinking and reasoning. Effective questions are planned before the lesson and when asked should include everyone in the classroom. A good teacher should give the learners time to think through the questions and should avoid judging the learner's responses. The following are some of the steps the teacher should follow when asking questions:

1. Pose the question to the whole class and in a natural friendly and conversational manner. The teacher can repeat the question only when necessary or when unclear.
2. Give time for learners to consider the question and think about the answer. Give three to five seconds after asking a question.
3. Call on a name of the learner that volunteer to attempt the question.
4. Listen to the answer from an individual learner and not chorus responses.
5. Follow up the learner's response in ways that encourage e.g. Comment, evaluate and reinforce the answer
6. Avoid asking a series of run-on questions while attempting to sharpen the focus of the original questions.
7. Sometimes allow the learners to ask you questions and before you respond you should give the other learners an opportunity to attempt in answering it. In case you do not know the answer promise to do it later after research.

E-tivity 3.2.3 How and When to Use Questions in the Classroom

Numbering, pacing and sequencing	3.2.3
Title	When and how to use questions

Purpose	The purpose of this e-tivity is to enable you to demonstrate when and how to use questions in the classroom instruction
Brief summary of overall task	Click here to watch the video and state several ways on when and how to ask good questions.
Spark	
Individual task	(c) Using bullet points, outline the best ways of asking good questions in the classroom (d) When do you think a teacher should ask questions when teaching?
Interaction begins	e) Select a topic in one of your teaching subjects and write some of the questions you are likely to ask your learners at the beginning, developing and summarizing your lesson.
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. ☐ Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	Reinforcement

3.2.4 Use of Reinforcement in the classroom instruction

The teacher is required to master the reinforcement skill in order to adopt appropriate techniques for giving feedback to the learners. The teacher needs the skill to be able to react to the learner's responses and their actions/ behavior in the classroom. Reinforcement is important because it modifies the learner's behavior. A teacher should react to a pupil's contribution to the teaching/ learning situation in such a way as to reinforce the event. This will encourage pupils to answer questions, make suggestions and participate in class discussions. There is

Close relationship between questioning and reinforcement and in the classroom situation they cannot be used separately.

Definition of Reinforcement

It is a form of learning whereby the contingency between a specific behavior and a desirable consequence help increase the likelihood of behaviour recurring (B.F. Skinner). Reinforcement is an action that make a target behaviour more likely to occur in future. Reinforcement can impact either positively or negatively to a learner and therefore as a teacher you should use reinforcement carefully and wisely.

Purposes of reinforcement skill

1. To increase the likelihood that a positive behavior will recur.
2. Increase learning.
3. Increase learners participation in learning activities.
4. Boost character development.
5. Contributes to self-esteem of the learners.
6. Leads to heightened enthusiasm in learners and even the teacher.

Types of Reinforcement

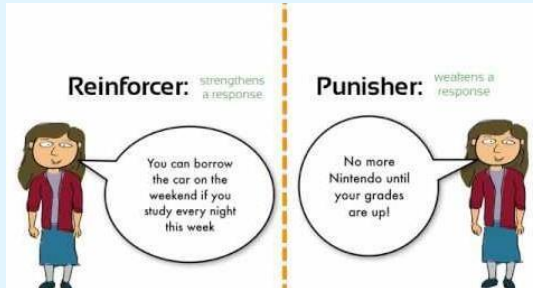
1. Positive verbal reinforcement. The teacher uses verbal expressions such as good, right, excellent, well done, go ahead, fine, correct, keep it up among others.
2. Positive non-verbal reinforcement. The teacher use gestures or some other behaviors like nodding the head, smiling, patting, moving towards the learner, friendly look, keeping eye contact, writing the response on the chalkboard etc.
3. Negative verbal reinforcement. The disapproval words are used like wrong, incorrect, not right, no and at times sarcastic remarks.
4. Negative non-verbal reinforcement. The teacher disapproves the learner's response without using words. Examples include frowning, staring, looking angrily, shaking the head etc.
5. Denial of reinforcement. No attempt of reinforcement in situations that warrant. This is likely to discourage the learners resulting to minimal participation in the learning process.
6. Delayed reinforcement, the emphasis on the positive aspects of learners' response by redirecting class attention to earlier contribution by a learner.

When and How to Use Reinforcement in the Classroom

The teacher should use reinforcement skills to all the responses from the learners and therefore the nature of words should be selected wisely to lessen the effect. This should be used in all the phases of a lesson i.e. introduction, development and conclusion. The reinforcement should consistently be done according to the planned schedule and delivered straight away after the learner's response. All in all, the teacher should consider the following components of reinforcement in the classroom instruction:

1. Acceptance- The teacher gives some indication that the learners contribution is acceptable or that he is at least listening to it
2. Establishing relationship - The learner's idea is conveyed to the class and a discussion is initiated. The teacher may formulate the idea more clearly or he may ask the learner to explain
3. Use- A contribution from one learner is used to stimulate discussion. The contribution may be used to solve a problem
4. Praise - The teacher encourages, praises or makes positive remarks about the learner's contribution. Praise should be honest and not flattery.

E-tivity 3.2.4 Use of Reinforcement in the Classroom

Numbering, pacing and sequencing	3.2.4
Title	Use of reinforcement in the classroom
Purpose	The purpose of this e-tivity is to enable you to demonstrate how to use reinforcement in classroom instruction
Brief summary of overall task	Click here to read the content and identify different types of reinforcement in article Click here and give examples of reinforcement used in the classroom
Spark	
Individual task	(e) Using bullet points, outline any four types of reinforcement a teacher can use in the classroom (f) Give examples of rewards and punishments a teacher can use during the teaching and learning process.

	(g) When and how should you use reinforcement skill in the classroom
Interaction begins	f) Demonstrate how you can use reinforcement in the classroom g) Explain some of the advantages of using reinforcement in the classroom.
E-moderator interventions	☐ Ensure that learners are focused on the contents and context of discussion. ☐ Stimulate further learning and generation of new ideas. ☐ Provide feedback on the learning progress. ☐ Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	Skill practice

3.3 Assessment Question

Choose a topic in one of your teaching subjects and then:

- Prepare a lesson plan for a lesson that will be taught through questioning skill. (10 marks)
- Write some five (5) questions you are likely to ask the learners in the lesson you have prepared above. (5 Marks)
- Indicate how you will reinforce learners' responses in case you receive a correct, partially correct or a wrong answer. (5 Marks)

3.4 References

Berner, S., C.G. Simpson and C. A Rose, (2012) Positive and Negative Reinforcement in Increasing Compliance and Decreasing Problematic behavior. *National Teacher Journal*, vol.5 pg.45-51

Rose, Sari, and John Litcher (1998) "Effective Questioning Techniques: In Theory and Practice." In *Studies in Teaching*, ed. Leah P. McCoy. Winston Salem, NC: Wake Forest University.

Wilen, William (1987 *Questions, Questioning Techniques, and Effective Teaching*. Washington, DC: National Education Association.

LESSON 4

SET INDUCTION AND STIMULUS VARIATION

4.1 Introduction

The word “Set” can be defined as a temporary, but often recurrent condition of a person that orients him/her towards certain environmental stimuli or events rather than towards others.

“Induction” simply means ‘introduction’. Thus, in ‘set-induction, we are saying or doing (or both) specific things prior to a learning situation which will direct the learner’s attention to the task in hand. The activities preceding learning will have an influence on the outcomes of the task, and some sets are more successful than others in achieving planned outcomes. On the other hand, “Stimulus Variation” means something that stimulates or acts as an incentive. Variation means the act of deviating from the norm standard, e.g. changing the voice, pitch, volume or performing an act that is different from the norm. The purpose of this lesson is to understand and apply the concept of ‘Set-Induction’ and Stimulus Variation in teaching and learning process.

4.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- 4.2.1 Define the concept of Set-Induction and Stimulus Variation.
- 4.2.2 Use various types of Set-Induction and Stimulus Variation in the classroom.
- 4.2.3 Manifest the use of Set-Induction when introducing a topic in the classroom.
- 4.2.4 Use different stimuli when teaching, especially a double lesson.

4.2.1 The Concept of Set-Induction and Stimulus Variation

Set-Induction is a technique used by a teacher to attract the attention of all the learners with a view to arousing their interest in the lesson. On the other hand, Stimulus Variation is a technique of teaching and refers to those teachers’ actions which develop and maintain a high level or maximum attention on the part of the learners in the course of the lesson, e.g. mannerisms, voice, personality, media and materials used during introduction of a topic.

Functions of Set-Induction

1. Focusing learners' attention on what is to be learnt by drawing their attention.
2. Acts as a means of transition from the familiar to the new, i.e. from known to the unknown, usually by using questions and answers at the beginning of a lesson or topic.
3. To provide a framework or structure for a lesson, for example, learners learn better when they are told in advance, or at onset what the teacher expects of them.
4. To give meaning to a new concept or principle. This often involves the use of concrete and specific examples and analogies to assist learners in comprehending abstract ideas and concepts.

4.2.2 When to Use Set-Induction

In addition to its use at the beginning of a lesson, set induction may also be used during the course of a lesson, for example, to:

1. Begin a new unit of work,
2. Initiate a discussion,
3. Introduce an assignment,
4. Prepare for a field excursion,
5. Prepare for a practical session in the laboratory,
6. Prepare for viewing a film or video, or listen to a radio programme,
7. Introduce a guest speaker.

In conclusion, Set-Induction is a technique used by a teacher/speaker to attract the attention of all the learners or audience to arouse their interests. Induction means initiation of knowledge to a beginner through inducement (refer to your first day in form one or the first day in the University and how you were inducted there).

4.2.3 Stimulus Variation:

Stimulus is simply something that acts as an incentive or factor that is able to cause a response in an organism, e.g. drugs, electrical impulses, among others. In varying the stimulus, the teacher aims to deviate from the sustained uniformity of presentation that can otherwise lead to boredom and mental inactivity. This is based on research evidence which indicates that changes in perceived environment attract attention and stimulate thought. The following are four main means of varying the stimulus:

1. **Teacher movement:** These are deliberate and timed shifts about the room which can help to revive and/or sustain interest. However, one should avoid nervous and irritating movements such as obsessively pacing up and down the same part of the room.
2. **Focusing behaviors:** Communication can be aided by the use of verbal focusing (giving emphasis to particular words, statements or direction) and gestural focusing (using eye movements, facial expressions, and movement of the head, arms and body). Gestures are important as means of communication between the teacher and the learner if being used to gain attention and express emotions. Verbal-gestural focusing, which is a combination of the two, can also be useful.
3. **Changes in Speech Patterns:** This involves changing the quality, expressiveness, tone and rate of speech, all of which can increase animation. Planned silences and pauses can also be effective.
4. **Changing interaction:** This is to bring about the changes on the main types of interaction; namely,
 - (i) The teacher and the class (the teacher lectures or demonstrates to the whole class and puts a question to the whole class);
 - (ii) The teacher and a learner (the teacher asking a question to a specific learner), and
 - (iii) A learner and other learner/s (the teacher redirects a learner's question or response to other learners).
5. **Shifting Sensory Channels:** Information is processed by means of the five senses and research suggests that learner's ability to take in information can be increased by appealing to sight and sound alternately. Thus, a teacher will follow up a verbal explanation with an accompanying diagram

E-tivity 4.2.3 Concept of Set-Induction and Stimulus variation

Numbering, pacing And sequencing	4.2.3
Title	Definition of the terms Set-Induction and Stimulus Variation

Purpose	The purpose of this E-tivity is to enable you to explain the meaning of Set-Induction and Stimulus Variation
Brief summary of the overall task	Go to Youtube, type “Set-Induction and Stimulus Variation”. Watch videos on Micro teaching by Fatima, M.Qasim, Fatin Syahirah Zubaidi, Rashvyne Chaal and Mira Sya. Also watch other videos on Stimulus Variation by Jerald Antony Raj, Dr Nisha Verma, Binod Kumar and Dr. Gajanan Gulhane. Go to https://images.app.goo.gl/pQx5pxJR Briefly explain the importance of Set-Induction and Stimulus Variation in teaching and learning
Spark 1	Use Set-Induction and Be smart in class. An example of a boring lesson lacking varied stimulus. Avoid this.   
Individual task	a) In your own words, define Set-Induction. b) Post three examples of Stimulus Variation used in classroom. c) Explain why Set-Induction and Stimulus Variation are important in teaching and learning.
Interaction begins	d) State two functions of Set-Induction and two of Stimulus Variation. e) Explain what you would do if you found out that some students tend to dose off during your lesson.
E-moderator interventions	• Make sure that the learners are focused on the contents and ask questions for Clarification if necessary • Encourage further reading and viewing of videos on Set-Induction and Stimulus Variation and other videos on Micro Teaching. • Provide feedback on the learning progress • Close the E.tivity
Schedule And time	This task should take 30 minutes
Next	Guiding Principles of Stimulus Variation

4.3 Assignment

- Using a topic from one of your teaching subjects, prepare a 40minutes lesson plan to illustrate aspects of Set-Induction and Stimulus Variation skills. (10 Marks)

2. Write some ten (10) actions you would perform to sustain your learners' attention and interest in classroom proceedings.
(10 Marks)

References

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Sniper View (1975) – The Australian Journal of Education–Volumes 19-20, page 121

Phil Race (2010) –Making Learning Happen; A guide for Post Compulsory Education.

Kate Exley, Reg Dennick (2004) – Giving a lecture – From Presenting to Teaching.

LESSON 5

SMALL GROUP WORK AND USE OF EXAMPLES

5.1 Introduction

In this lesson, we will define small group work and discuss how it can be used in teaching and learning process. We will also discuss how as teachers we can use examples to illustrate, clarify and simplify the concepts being taught. Small group work is a strategy that requires learners to engage in learning activities for specified period of time. It involves learners working collaboratively on either same or different tasks in the classroom. When learners work in groups they are likely to be involved in thinking and doing things together. It is important for a teacher to think through the nature of learning task or activity to engage the learners in order to determine the duration of time to be spent in the group activity. In teaching, the teacher should also use several examples to make the teaching and learning process to be more interactive and for the process to be meaningful. The purpose of this lesson is to enable the learners to understand how they can use small group work and examples in the classroom instruction.

5.2 Lesson Learning Outcomes

5.2.1 Identify the types of groups that can be used in the classroom instruction

5.2.2 Demonstrate how to use small group work

5.2.3 Demonstrate how to use examples in the classroom instruction

5.2.1 Types of groups

Groups can be formed either heterogeneously (different characteristics) or homogeneously (same characteristics). In the classroom situation the teacher may decide to form groups through random sampling, natural socialization, learners' intellectual abilities, gender and competencies. It is advisable for the teacher either to assign the learners groups, allow the learners to choose groups or use randomized group work. There are four types of groups a teacher can use in the small group work and they are as follows:

1. Buzz groups

The learners discuss a question or problem in twos or threes according to their sitting arrangement. In this group the learners share and exchange ideas, opinions and suggestions. The task is very simple and may be accomplished in less than five minutes. The learners discuss the same question in all groups in the class. A teacher may randomly select few groups to share their points/responses which could be used to develop the lesson further.

2. Task groups

The grouping is an extension of buzz groups where the number of participants may increase but not exceed five. The task is the same across all the groups and is more detailed, demanding and time duration is longer than five minutes. The group organization is formal with a leader to coordinate the group activities and the secretary to record the points of discussion. The reports from the groups are exhaustive as each group may discuss the task/activity in different ways or on a different angle of the same task. The teacher may randomly select few groups to report which may stimulate discussions in the classroom.

3. Syndicate groups

Syndicate group is composed of five members with a leader and secretary. Each group is given a different task to discuss. The time duration in the group is longer more than ten minutes and therefore the teacher should be creative enough to come up with different activities. Each group is given time to report its findings. The teacher uses the information to develop the lesson, clarify the content and draw conclusions.

4. Panel grouping

Panel group involves the teacher selecting a few members to come in front of the class and they are given a task or an issue to discuss in different perspectives. Each member approaches the issue in a different way that may lead to a debate in the classroom.

E-tivity 1.2.1 Types of groups

Numbering, pacing and sequencing	5.2.1
Title	Types of groups
Purpose	The purpose of this e-tivity is to enable you to identify types of groups used in the classroom instruction
Brief summary of overall task	Click here to read the article and describe types of groups used in the classroom instruction

Spark	
Individual task	(a) Using bullet points, explain any four types of small group work used in instruction (b) When do you think a teacher should use group work in instruction?
Interaction begins	c) Select a topic in one of your teaching subjects and write some of the group activities that you are likely to engage your learners in..
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. □ Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	How to use group work instruction

5.2.2 How to use group work

The teacher should follow the following steps in introducing the group activity:

1. Share the rationale for using group work.
2. Have the learners form groups before giving them instructions?
3. Facilitate some form of group cohesion-make sure the learners introduce themselves in a group and build some trust.
4. Explain the activity/task clearly.
5. Set ground rules for group interaction.
6. Allow the learners ask questions/ seek clarifications or raise concerns before embarking on the group activity/task.

Preparing and monitoring the group work

The following should be considered by the teacher when preparing group work

1. Think carefully about how the learners will be arranged in groups in the classroom by reflecting to the following questions; will the learners be comfortable with the groups formed? How about the classroom layout?

Will the learners be able to hear each other? Would I be able to monitor the groups?

2. Group etiquette. The learners should be able to respect others opinion, contribution, differences and create an inclusive environment.
3. Talk to the learners about their past experiences with group work and allow them to come up with their ground rules.
4. Monitor the groups but not interfere with the group functioning. You could move around the groups and listen to the discussion.
5. Expect a lot from the learners by expressing your confidence in them as you move around the classroom.
6. Be slow to share what you know. Allow the learners to do the activity on their own until they accomplish.
7. Make the learners understand your role as a facilitator.
8. Provide closure to the group discussion by ending with a plenary where by the learners must do group reporting either orally or through written reports.
9. Clarify on how you want the learners to participate.
10. Link the ideas raised by the learners to the content and the lesson outcomes.
11. Allow the learners to reflect on the group work process.

Advantages of group work

1. Improve understanding through sharing of ideas
2. Stimulates self-expression, listening and reasoning
3. Can lead to development and emergence of leadership qualities
4. Sharing of ideas when there are limited resources
5. Development of interest due to practical involvement
6. Fosters cooperation and reduces individual frustration
7. Gives learners opportunities to practice the democratic process
8. Practice of oral communication and enhancement of other skills

Limitations of small group work

- It is time consuming
- Not convenient for large classes
- Assessment of individual contributions is difficult
- Difficult to choose suitable tasks and to form groups

E-tivity 5.2.2 how to use small group work

Numbering, pacing and sequencing	5.2.2
Title	How to use small group work
Purpose	The purpose of this e-tivity is to enable you to demonstrate how to use small group work in classroom instruction
Brief summary of overall task	Click here to read the chapter and then click video to watch and explain how different small groups are used for classroom instruction.
Spark	
Individual task	- Using bullet points, outline the best ways of using small group work in teaching and learning process. - What are the merits and demerits of using small group work in teaching?
Interaction begins	- Select a topic in one of your teaching subjects and write some of the group activities that you are likely to involve your learners. - Explain the strategies you will come up with when forming groups in the classroom
E-moderator interventions	- Ensure that learners are focused on the contents and context of discussion. - Stimulate further learning and generation of new ideas. - Provide feedback on the learning progress - Mediate group to group participants - Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	Use of examples

5.2.3 Use of Examples as a Teaching Skill

Examples are used by teachers to simplify and clarify the ideas or concepts being taught. At times the concepts may be difficult or complex and for the learners to

understand them, the teacher should use a variety of examples or illustrations. For teaching to be meaningful, the teacher should use simple examples to link the prior knowledge to the new knowledge, relevant examples to the content being taught and interesting examples that arouse the learner's curiosity and interest.

How to use examples

1. Examples should be prepared in advance.
2. The teacher should be observant of learners' behavior and their verbal responses to be certain the examples are appropriate.
3. Sufficient number of examples must be provided while at the same time a variety of situations in which the rules / concepts may be observed are indicated.
4. There should be a clear and unequivocal link between the example given and the relevant rule, principle or concept.
5. The teacher should always encourage the learners to give additional examples.

Approaches to Using Examples.

1. Inductive i.e. moving from a particular example to the generalization or starting with the examples then giving the generalization based on the given example.
2. Deductive i.e. moving from the generalization to the examples. Starting with the rules then giving examples to further illustrate.

E-tivity 5.2.3 use of examples

Numbering, pacing and sequencing	5.2.3
Title	Use of examples
Purpose	The purpose of this e-tivity is to enable you to understand how to use examples in the classroom instruction
Brief summary of overall task	Click video and watch an example of a lesson with illustrations and examples.
Spark	
Individual task	(a) Using bullet points, explain why teachers use examples and illustrations in teaching (b) Discuss some of the ways a teacher should consider when using examples or illustrations in teaching and learning process

Interaction begins	c) Select a topic in one of your teaching subjects and write some of the examples and illustrations you are likely to use when teaching it
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. ☐ Close the e-tivity
Schedule and time	This task should take 30 minutes
Next	Practice of the skills

5.3 Assessment Question

1. Choose a suitable topic in one of your teaching subjects and prepare a lesson plan showing:
 - (a) The use of group activity.
(10 Marks)
 - (b) Illustrate your lesson with the use of suitable the examples.
(10 Marks)

5.4 References

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LESSON 6

INTEGRATED SKILLS

6.1 Introduction

This is a skill used by the teacher in teaching and learning and which focuses on the four main skills of reading, writing, speaking and listening through a communicative language teaching methodology. In this skill, learners are engaged in various activities, such as, role-playing, discussions, listening, writing and speaking, among other activities in the classroom. The purpose of this lesson is to ensure that you will understand and apply this skill in delivering your content in the classroom.

6.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

6.2.1 Define integrated skill as used in teaching and learning in the classroom,

1.2.2 Demonstrate integrated skill in your teaching lesson,

6.2.3 Use integrated skill effectively in the classroom.

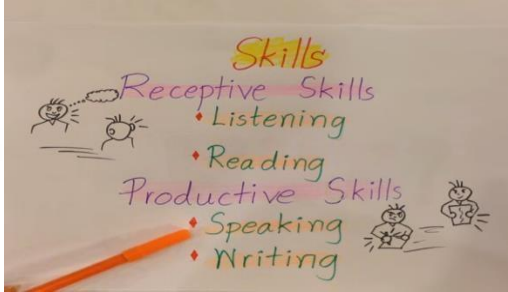
6.2.1 Concept of Integrated Skills

Integrated skill is divided into two parts namely, receptive skill and productive skill. Receptive comprises listening and reading, while productive comprises speaking and writing. In this skill, learners will be engaged in various activities to practice what they are learning during the lesson.

These activities include listening to the teacher's explanations, observe his demonstrations, roleplay, discuss in groups, ask and answer questions among other tasks. This is a skill where the teacher uses various skills in delivering the planned content in class.

6.2.1 E-tivity: Concept of Integrated Skills

Numbering, pacing And sequencing	1.2.1
Title	Definition of the term Integrated Skills in teaching and learning.
Purpose	The purpose of this E-tivity is to enable you to define and explain the meaning of the term Integrated Skills in teaching and learning.

Brief summary of the overall task	Go to www.pdst.ie/publications and www.pdst.ie/pedagogy for better understanding of the skill. Make your own notes. Also, open and watch YouTube video by John Mark King, lesson 1 & 2. 2. From the videos you have watched, briefly explain the advantages of using integrated skill in teaching and learning.
Spark 1	  INTEGRATED SKILL AT ITS BEST FOR A COURSE AN EXAMPLE OF USING INTEGRATED SKILL IN TEACHING COMMUNICATION SUBJECTS
Individual task	(a) Define the term “integrated skill”. (b) In your own words, explain the importance of using “integrated skill” in teaching and learning. (c) Using one of your teaching subjects, name a topic in which you can use integrated skill.
Interaction begins	(d) State two components of integrated skill. (e) Explain two different activities that you would give to shy students to instill confidence in expressing themselves. (f) Give at least three advantages of using integrated skill in teaching and learning.
E-moderator interventions	• Keep the learners focused by posing short questions. • Encourage them to use their own words when defining terms. • Give feedback by answering any of the questions on the lesson. • Assign further readings. • Close the E-tivity.
Schedule and Time	This task should take 50 minutes.
Next	Merits and demerits of using integrated skills in teaching and learning.

6.3 Assignment

1. Research and read more on the use of the integrated skills in teaching and learning.
2. Write a lesson plan for 40 minutes illustrate the integrated skills.

6.4 References

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LESSON 7

WRITING AND INTERACTIVE BOARDS

7.1 Introduction

In this lesson we will discuss different types of writing and interaction boards. Researchers argue that the most iconic tools used in teaching and learning process are writing and interactive boards. Learners learn better by having information presented through visual means and the boards are the simplest and easiest visual teaching tools. The purpose of this lesson is to enable the learners to understand the importance of using writing and interactive boards in teaching and learning process.

7.2 Lesson Learning Outcomes

By the end of this lesson you will be able to;

7.2.1 Define the term writing and interaction boards

7.2.2 Identify different type of interaction boards

7.2.3 Demonstrate how to use writing and interactive boards

7.2.1 Definition of writing and interaction boards

Writing boards are boards that can be written on by touching it with a figure, or using a pen or chalk. It can be electronic or board made from wood. Writing or interactive boards enhances classroom instruction by engaging the learners in individual or group solving problems and also eliciting learners' ideas through class brainstorming. Boards are very important because of their inherent accessibility, versatility and adaptability. They be used by both teachers and learners in lectures, flipped classes, tutorials, and even in seminars to present the content, expand information, draw diagrams, illustrations, display learning resources like charts, maps etc.

E-tivity 7.2.1 writing and interactive boards concept

Numbering, pacing and sequencing	7.2.1
Title	Writing and interactive boards
Purpose	The purpose of this e-tivity is to enable you to understand the meaning of writing and interactive board
Brief summary of overall task	Click here to read the article on interactive boards

Spark		
Individual task	a) Explain the meaning of interactive boards b). Discuss the advantages and the disadvantages of using writing and interactive boards in classroom instruction.	
Interaction begins	c) Post some of the advantages and disadvantages of using writing and interactive boards in teaching and learning process.	
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. □ Close the e-tivity	
Schedule and time	This task should take 20 minutes	
Next	Types of writing and interactive boards	

7.2.2 Types of writing and interactive boards

There are various types of writing and interactive boards that can be used in the classroom instruction. Some of them includes:

1. Chalkboards where chalk is used
2. White boards where pens are used
3. Smart boards that is electronically enabled etc.

E-tivity 7.2.2 Types of writing and interactive boards

Numbering, pacing and sequencing	7.2.2
Title	Types of writing and interactive boards
Purpose	The purpose of this e-tivity is to enable you to identify types of writing and interactive boards.

Brief summary of overall task	Click video to watch different types of writing and interactive boards
Spark	
Individual task	a) Using bullet points, outline different types of writing and interactive boards b) Mention any other type of the writing boards that has not been discussed. c) Which types of writing and interactive boards can you feel comfortable using in your teaching activities and why?
Interaction begins	d) Post some of the types of writing and interactive boards you think you can use in your teaching e) Explain the reasons for choosing the types of interactive boards above.
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress • Mediate group to group participants • Close the e-tivity
Schedule and time	This task should take 20 minutes
Next	How to use writing and interactive boards

7.2.3 How to use writing and interactive boards

The writing and interactive boards can be used at the beginning of the lesson, during the lesson and after the lesson. The boards may be used for different purposes like writing complex concepts, examples, main points of the content etc. the following points should be considered when using writing and interactive boards;

1. Plan on how you are going to utilize the boards in advance
2. Neatness and tidiness are very important
3. Write using large or legible handwriting
4. Write in a straight line

5. Highlight key words by either bolding, underlining or using a different colour
6. Erase irrelevant information as you write
7. Clean the board after use.

At the beginning of the lesson, divide the board into sections depending on your plan. One section could be used for examples, the Centre section the main points and the other section for diagrams, illustrations or for displaying charts, maps etc. The teacher should write the subject, date topic and lesson outcome on the board so that the learners to know where the teaching and learning process is headed.

In the development stage of the lesson, the teacher or instructor should use the sections divided appropriately. The teacher should avoid facing the board at the same time talking to the learners. Facing learners makes them pay attention to the lesson and engaging them is easier. The two should be done differently. The teacher should write and step aside to allow the learners to copy from the board, digest the material and even allow them to comment ask questions. In case the teacher expects to draw a diagram, it should be done carefully and humorously. In case a teacher makes a mistake on a board, it is advisable either to erase the word and write it a fresh or cross it out and write the information correctly.

At the end of the lesson, the teacher should erase the board and if there is some information that need to be reserved for later used, this should be made clear.

E-tivity 7.2.3 How to Use Writing and Interactive Boards

Numbering, pacing and sequencing	7.2.3
Title	How to use writing and interactive boards
Purpose	The purpose of this e-tivity is to enable you to demonstrate how to use the writing and interactive board
Brief summary of overall task	Click here to read about how to use writing and interactive boards

Spark	
Individual task	a) Discuss how you can use any writing and interactive boards in your classroom instruction from the beginning of the lesson to the end.
Interaction begins	b) Post a sample of the activities on your board from the beginning to the end of the lesson.
E-moderator interventions	• Ensure that learners are focused on the contents and context of discussion. • Stimulate further learning and generation of new ideas. • Provide feedback on the learning progress. ☐ Close the e-ivity
Schedule and time	This task should take 20 minutes
Next	Writing and interactive board practice.

7.3 Assessment Question

1. Identify any three writing and interactive boards and explain how you can use them in your classroom instruction. **(10 Marks)**

7.4 References

Davis, B. G. (2009). *Tools for teaching*. John Wiley & Sons.

Karsenti, T. (2016). The Interactive Whiteboard: Uses, Benefits, and Challenges. A Survey of 11,683 Students and 1,131 Teachers. *Canadian Journal of Learning and Technology* 42.5: 1-22.

LESSON 8

INSTRUCTIONAL GRAPHIC MATERIALS

8.1 Introduction

Graphic instructional materials or simply graphics are teaching/learning materials that use a combination of drawings, words and pictures to communicate facts and ideas clearly and succinctly. This is illustrated by the saying, “a picture paints a thousand words”.

Graphic materials are among the most useful and versatile teaching aids at the disposal of a teacher. They enable the learner grasp information and concepts quickly and effectively. This is because as revealed by research, the sense of sight contributes an overwhelming eighty percent (80%) to learning. By contrast, hearing contributes to ten percent (10%), and the other senses; namely touch, taste and smell- each contribute about three percent (3%). Clearly then, the more we engage the sense of sight in the learning process, the greater will be the learning. One important way of engaging the sense of sight is through the use of graphic materials.

The purpose of this lesson is to demonstrate the importance of graphic materials in the teaching and learning process and to provide you the skills that will enable you produce effective graphic materials relevant to learning of your teaching subject/s.

8.2 Lesson Learning Outcomes

By the end of this lesson you should be able to:

8.2.1 List the benefits of graphic instructional materials.

8.2.2 Describe the characteristics of the various types of graphic materials.

8.2.3 Outline some major considerations relevant to the production of effective graphic materials.

8.2.4 Explain the criteria of good graphics.

8.2.5 Make graphic materials to be used to effectively teach content relevant to your teaching subject/s.

8.2.1 Benefits of Graphic Instructional Materials

Under various circumstances, graphic materials will lend themselves to effective learning. This is because they can help to:

- (a) Visualize ideas and concepts- Graphic materials enable a teacher to effectively explain many complex and abstract concepts. For example, to explain the complex process of breathing in and out in animals via the synchronization of the lungs, ribs and the diaphragm.
- (b) Highlight important information. Graphic materials simplify concepts taught by presenting the essential/key information. For example, the types of stitches in home economics.
- (c) Summarize certain types of information and ideas- For example, statistical information can be presented visually in form of graphs and musical information in form of musical notes.
- (d) Motivate learning- whenever well-made graphic materials are used, the levels of interest in a lesson are heightened as the lessons are usually more concrete and realistic. This usually leads to greater engagement of the sense of sight and more learning.

8.2.2 Types of Graphic Materials

Graphic materials can be categorized into six groups:

1. Charts
2. Diagrams
3. Graphs
4. Posters Maps and
5. Cartoons and comics

1. Charts

A chart is a combination of pictorial, numerical or verbal information designed to present a clear visual summary of an important an important process or set of relationships.

Depending on the content presented in a chart, charts may be classified into the following types:

1. **Classification charts**- used to classify/categorize objects and events; such as parts of speech, taxonomy of animals, and so on.
2. **Flow/process charts**- show a sequence, procedure or as the name imply the flow of a process. These charts are usually drawn horizontally.
3. **Tabulation charts**- information presented in form of a table; for example, a timetable.
4. **Chronology charts**- show time relationships or historical sequence of events.

5. **Relationship/organizational charts**- show a chain of command in an organization as say a company or government department.

In addition to these types of charts, there is a special kind of chart- the [flip-chart](#). A flip-chart refers to a pad of large sized paper fastened together at the top. The individual sheets of paper each hold/present a limited verbal/visual message and are arranged for a sequential presentation to a group.

2. Diagrams/Drawings/Sketches

In general, a diagram is a more finished representation than a sketch. A diagram is a drawing illustrating the main features of an object, process or area.

3 Graphs

Graphs are visual representations of numerical data. They enable us to grasp quickly and accurately the meaning of masses of statistical information.

Forms of graphs include: Line, Bar, Circle/pie and pictorial graphs

Much of information often presented in tabulation charts can be converted into graphs. Data can be grasped or interpreted more quickly in a graph than in a tabular form. Graphs are also more visually interesting.

4. Maps

A map is a flat representation of some or the earth's entire surface on a scale.

Every effort should be made to produce "readable" maps. This may be achieved by avoiding irrelevant details; matching the size of the map to its function; making good use of color and relating symbols and color to objects presented.

Maps may be used to show surface features; the directions and distances of places; scientific data such as ocean currents; political boundaries; socio-cultural factors such as populations, language patterns, political boundaries; economic data and the like.

5. Cartoons and comics

A cartoon uses satire, caricature or exaggeration to represent pictorially ideas, persons, groups or situations.

Cartoons are serious in intent and have the power to capture attention and influence attitudes and behavior. Most cartoons have characteristics that include minimal

detail, use of familiar symbols and characters and stereotypes that are quickly recognized and understood. The message presented is usually crispy and clearly communicated.

In social studies, cartoons have proved to be an effective means of motivating current events discussions and examination of different political views. Valuable learning results from students drawing their own cartoons.

A comic, on the other hand, is a form of cartoon in which characters enact a story in a sequence.

6. Posters

A poster is a visual combination of bold design and color and/or a brief written message which is meant to instantly catch attention and convey the message.

Usually each poster deals with one idea at a time. It must also be clear and large enough to be read from a good distance. A poster can be used to stimulate or motivate interest in a new topic or unit.

7. Note

Besides the six (6) types of graphics discussed above, you may also include pictures and photographs. Sometimes maps, charts, graphs and so on are all referred to as charts. You will also notice that there is a certain amount of overlap between the various types of graphics. It may not be quite distinct for example, where a diagram begins and where a chart starts.

Taken as a whole, all graphics materials seek to principally communicate information primarily through the visual channel. Any written information should merely assist furthering the visual information and NOT the reverse. This means that little or no writing is admissible.

8.2.3 Some Production Considerations

The effectiveness of any graphic material is largely dependent upon its production quality. To produce effective graphics, you should attend to the following three aspects:

1. Composition/layout
2. Lettering
3. Illustration

1. Composition/Layout

- (a) Select appropriate size of working area. The working area should be large enough to contain information to be seen by the whole class.
- (b) Consider whether the information will be better presented with the working surface positioned horizontally or vertically.
- (c) Use as much of the working area as possible, leaving only a reasonable margin around the area.
- (d) Choose for presentation only the major points, since only few points can realistically be shown.

2. Lettering

- (a) Ensure the heights of the letters are consistent.
- (b) Use simple and bold lettering style.
- (c) Use color or underline for emphasis.

3. Illustrations

Illustrations can be varied. They may be photographs, drawings, paintings, diagrams, maps, specimens, and so on. Pay attention to the following as you choose your illustrations:

- (a) The quality of illustrations will depend largely on their clarity and suitability for the intended purpose.
- (b) Drawings and diagrams should be simple and bold.
- (c) Make use of color; it adds realism as well as attraction to the work.
- (d) Select harmonious and effective color.
- (e) Enlarge/reduce pictures accordingly to suit the intended work.

E-tivity 8.2.3 Production Considerations

	8.2.3
Title	Production of E-graphics
Purpose	The purpose of this e-tivity is to enable you to produce graphic materials that would be useful in promoting learning.
Summary of task	Watch video1 and video 2 on how to develop graphic material from the internet for instructional purposes.

Spark	
Individual task	(a) As you watch the video, list the key steps in transferring a diagram/graphic material to your work space. (b) Choose a picture/diagram relevant to content in your subject area and with appropriate modifications prepare a graphic instructional material for a class of your choice.
Interactions	(c) Post the key steps in transferring graphic materials from the internet. (d) Encourage all to practice the skills of producing graphic instructional materials from the internet.
E-moderator interventions	• Review the key steps in transferring graphic materials from the internet. • Provide positive and constructive feedback on responses received. • Close the activity
Schedule and time	This task should take one hour
Next	The criteria of good graphics

8.2.4 Criteria for Good Graphics

A good graphic material should be:

- (a) Clear in that, the production technique brings out the message clearly.
- (b) Precise in that, the message is straight forward.
- (c) Accurate in that, the information presented is authentic in all respects.
- (d) Simple in that, it is not loaded with unnecessary details.
- (e) Bold in that, it is big enough to be seen from an appropriate distance.
- (f) Readable in that, it can be interpreted by the intended audience.
- (h) Interesting in that, it is well made, attractive to look at and tidy and well preserved.

8.2.5 Presentation and Storage of Graphic materials.

Charts often lose their impact unless stored and presented in the right way and order.

- (a) It is important that a chart is exposed to learners at just the right moment in the lesson to avoid premature learner attention to information.
- (b) Mount the chart on the blackboard, a wall or an easel to ensure it is visible to all learners.
- (c) Make use of a pointer as you explain the chart.
- (d) Require reaction of your learners to gauge that they are making correct interpretation of the chart by asking pertinent questions.
- (e) Useful learning is gained by requiring learners to make drawings and diagrams.
- (f) As a general rule, graphic materials should be stored flat in large folders. They should NOT be rolled or folded.

8.3 Assignment

- (a) Adhering to the criteria of good graphics, produce any one of the graphic instructional material discussed in this lesson. The content covered on your graphic material should be relevant to any one of your teaching subjects.
- (b) Accompany your work with a lesson plan that clearly shows **what for?** **How?** and **when?** the graphic material will be used in the course of the lesson.
- (c) Credit will be given for quality, relevance and creativity.

8.4 References

ICT for teachers

https://www.youtube.com/results?search_query=ict+for+teachers%3A+how+to+use+microsoft+word+to+draw+a+diagram

ICT for Teachers, <https://www.youtube.com/watch?v=dqdyW-l2eVc>

LESSON 9

AUDIO- AND AUDIO-VISUAL PRODUCTION

9.1 Introduction

The process of producing audio recordings can be discussed under three major stages:

1. Planning
2. Script writing
3. Presentation / recording

9.2 Lesson Learning Outcomes

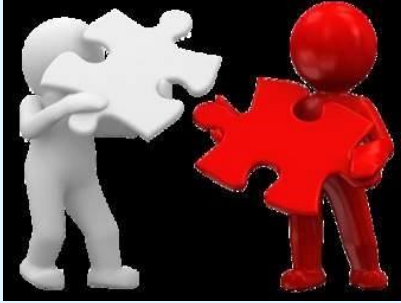
By the end of the lesson, you should be able to:

- 9.2.1 Plan a lesson
- 9.2.2 Prepare a program outline
- 9.2.3 Write a script for the lesson
- 9.2.4 Record and edit the recording
- 9.2.5 Operate audio-visual equipment
- 9.2.6 Locate sources of audio and audio-visual materials

9.2.1 Planning

The planning stage starts with the selection of topics to make up the series. A series of topics for a particular class is made of topics enough to cover a year. In some countries a series is made up of 24 audio lesson topics namely eight per term. The selection is guided by the specific functions of the medium as well as the needs of the subject or course. We must bear in mind the weaknesses and strengths of the audio / television medium as we select these topics. We must also give priority to topics that are not well covered in the class textbooks as these are the topics that actually need the audio support. The need arises due to scarcity of instructional materials and the nature of the content. Audio recordings can for example be useful in the teaching of music where the class teacher cannot demonstrate some of the tunes in the classroom or in the teaching of literature where different scholars give different interpretations of the same novel. Throughout the selection the writer must consult with the subject specialist.

E-tivity 9.2.1 Planning

Numbering, pacing and sequencing	9.2.1
Title	Planning
Purpose	The purpose of this e-tivity is to enable you to be able to select a sub strand that may need audio support.
Brief summary of overall task	Read the content in the notes under section 9.2 and make brief notes on the planning process.
Spark	
Individual task	After Reading material in the notes: - Recalling the five senses you learnt in your earlier courses, discuss the main weaknesses and strengths of using the audio medium only for instruction - For a named strand in one of your teaching subjects, identify at least two sub strands that may benefit from audio support. Justify the choice of the sub strands. Describe the nature of the support. - Save the work in your portfolio
Interaction begins	- Using short clear sentences, discuss three reasons why you would choose audio media over other media. - Post your work in the discussion forum - Read what your colleagues have posted - Read comments from your facilitator to get feedback - Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This task should take one hour
Next	Preparing programme outline (Teachers' notes)

9.2.2 Preparing Programme Outline (Teachers' notes)

The next step under the planning stage is writing the programme outline for each topic. This outline is made up of six elements. The first element is the programme title which must be concise and precise. For example the titles “Coffee growing” and “Coffee growing in Kenya” are different since in the former, the programme cannot confine itself to coffee growing in one country only or region but must address itself to coffee growing globally. The next element is the specific learning outcomes (objectives) we wish to achieve through the programme. These will depend on the class level the programme is intended for. It is important in writing these learning outcomes to cater for the different key performance indicators (domains of educational objectives). The outcomes must be achievable in the time available for the programme. The outcomes if well stated become very useful in content selection.

The third element of the programme outline is the content summary. Here the writer must now list the main facts, ideas, skills and attitudes to be covered by the programme. This content must be arranged in a logical manner for easy audio lesson development. The content summary if well prepared at this stage helps a great deal in deciding on the reference materials, resource persons to be interviewed and the illustrative examples if any. Once the content summary is in place it is then possible to decide on the fourth element of the outline namely form and structure. Deciding on the form and structure involves choosing whether to use straight talk, illustrated talk, interview, panel discussion, documentary techniques, dramatization, or a mixture of the above (magazine). The writer can then concentrate on the fifth element of the outline namely developing support materials to accompany the recording. The materials should encourage active participation of the learners in the programme. The material must involve learners at three critical stages of the lesson. The learning activities make up the sixth element of the outline. These stages are activities before the listening, during the listening and after the listening. Once the outline for the topics have been completed, it is advisable to discuss them with other subject specialists as to the validity and relevance of the proposed content.

The programme outline which comprises the sub strand, learning outcomes, content summary, support materials, and learning activities will be properly written out to make up the Teachers' notes. The Teachers' notes is the document that will guide the classroom teacher on how to make effective use of the audio recording.

SAMPLE TEACHER'S NOTES CLASS:

Standard 7

SUBJECT Home Science

TOPIC: Basic food groups for a balanced diet

Learning outcomes

By the end of the radio lesson, the learner should be able to: (i) Classify common foods into 3 food groups (ii) State functions of foods in each group.

Learning Aids

A chart showing drawings and mountings of the three food groups.

Activities before the broadcast

Pupils should be asked questions such as the following by the class teacher:

- (i) Name 3 good eating habits
- (ii) State at least 4 factors that influence eating habits
- (iii) Name 3 ways of obtaining food

The class teacher should also write the following on the chalkboard: -
Carbohydrates - energy giving foods

Proteins - body building foods

Vitamins - protective foods

Activities during the broadcast

The class teacher should receive responses from pupils when asked to do so by the radio teacher. S/he should quickly write down these answers. The teacher should display the chart showing drawings and mountings of the 3 food groups. S/he should point at the relevant parts when asked to do so by the radio teacher

Activities after the broadcast

The classroom teacher should ask pupils to write in their exercise books all the foods they eat at home and finally classify them. The pupils can also be asked to choose foods from these three groups and plan balanced meals for a whole week.

Once the script has been written and Teachers' Notes compiled, the script editor should edit the programme. The editor should consider the flow of content, the time slot allowed for the programme and the general layout of the script. He is to cross check the appropriateness of the recorded inserts and the music use. He also ensures that the signature tune chosen is appropriate. The signature tune is that piece of music or vocals that comes at the beginning and at the end of a programme. It is the same for a whole programme series and is regarded as the "trademark" of the programme series it identifies the programme. Usually the message in the signature tune is related to the theme of the series. The signature tune should never be used as bridge music. If the introductory remarks of a programme are to be voiced over the signature tune then the tune should be instrumental for it to be faded under the speech effectively.

E-tivity 9.2.2 Preparing programme outline (Teachers' notes)

Numbering, pacing and sequencing	9.2.2
Title	Preparing programme outline (Teachers' notes)
Purpose	The purpose of this e-tivity is to enable you to prepare the programme outline for a sub strand of your choice
Brief summary of overall task	Read the content in the notes under section 9.2.2 and be able to prepare the programme outline (Teachers' notes)
Spark	
Individual task	After Reading material in the notes: - In your subject area, identify a topic that in your opinion needs audio support. - After deciding on the programme level, identify the instructional outcomes and generate the content summary, list of resources and learning activities - Translate the programme outline into the Teachers' Notes. - Save the work in your portfolio
Interaction begins	- Using short clear sentences, describe the process of preparing the Teachers' notes. - Post your work in the discussion forum - Read what your colleagues have posted - Read comments from your facilitator to get feedback - Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This task should take one hour
Next	Scriptwriting

9.2.3 Scriptwriting

The art of scriptwriting involves laying on paper a simulation of how people really speak. The key point to remember is that the audio channel is for the ear and not for the eye. The sentences must be short, consisting of familiar phrases since we rarely use long sentences in everyday speech. Long rumbling sentences must be broken down into shorter ones. The active quality of natural speech ensures that the scriptwriter captures the attention of the listener and conveys information. The best way to write an audio script would therefore be to put down word by word the way you would tell the story to a friend. The words must sound warm and personal to compensate for the loss of visual communication.

The writer should consider whether the content suggests more than one presenter, a need for music, indigenous sounds, sound effects or simply one voice commentary. A good educational broadcast lesson should be voiced by a radio teacher and at least two radio pupils besides other presenters. It is good to have more than one presenter to give the programme some 'taste' but it comes but it becomes confusing when the voices are too many. Consideration should also be given to using sound effects at appropriate stages to give the programme a natural outlook for instance when the script is referring to an event taking place at dawn then the sounds of birds and other insects could bring in the time perspective. If the script is referring to an interview involving a poultry farmer, then during the interview sounds of hens and cocks should be heard in the background. The programme must open and close with a signature tune. This is a sound effect that identifies the programme series, it should where possible show some relation to the content or theme of the programme.

The following rules should be followed in script writing: -

- (a) The script must convey the message in a logical manner. It should be made up of one's thoughts arranged in a sequence to drive the listener to stay tuned.
- (b) All the facts must be factual, i.e accurate, honest, current, truthful and sincere.
- (c) The script must be 'mobile', interesting and relevant to the environment. It must be illustrative so that the listener is mentally visualized.
- (d) The script writer must have a complete story in him.
- (e) A good script must have a 'hook' which is catchy, to catch the listeners attention and sustain it throughout the listening session.
- (f) The script must avoid overloading the programme with lots of data or dates that would confuse the listener.

The script layout should be standard as this helps in ensuring that all those involved in the production understand its form and structure. An audio script should be made up of four distinct parts. The first part should be the introduction. Here the radio teacher should welcome the listeners to the programme and connect it to previous knowledge. The second part is the main body of the programme where the content should be delivered logically, punctuated with the pupil activities. The programme should allow for pauses to give pupils time to respond to questions and tasks. The third part of the script is the closing. Here the radio teacher with the

assistance of the radio pupils summarize the main points of the lesson. If necessary, the next topic can be stated and the pupils asked to collect necessary materials. A series of questions to direct the follow up can also be posed. Finally, some programmes finish off by giving credit to participants namely the script writer, script editor, the presenters, the producer, the technician and the station.

It is important to remember that the script should be:

- (i) Typed, double or treble spaced with margins of 1 ½ to 2 inches for easy reading
- (ii) With pages numbered sequentially
- (iii) With each page starting on a new sentence
- (iv) Reproduced on single sided non-flimsy paper
- (v) With enough copies for all those involved in the production (producer, presenters, and technician)

Materials to be recorded in the studio should be separated from pre-recorded materials (taped inserts for interviews, music and sound effects). Identifying details should be given for all inserts including name of interviewee, title of music, description of sound effects, cue-in (opening words or sounds) and cue-out (closing words or sounds) and exact duration .

SAMPLE SCRIPT

Programme series	: Primary Home Science
Programme title	: Basic food groups for a balanced diet
Programme level	: Standard seven
Programme number	: 4 of term 1
Script writers	: Raphael Muruu, Selina Ominde, Eunice Obat, Gordon Oluoch
Script-editor	: Lordvicus W. Olanga
Producer	: Eunice Obat
Technician	: Jacob M. Oloo
Duration	: 14 minutes 25 seconds SOUND
EFFECTS:	Signature tune
	: Bridge music

1. **CONTROL:** Bang on the signature tune...15seconds.... then fade under
2. **ANNOUNCER:** We present Home Science for standard seven. This is the fourth programme this term. The title of today's programme is "Basic food groups for a balanced diet". During the programme we will learn about: -
 - (i). The definition of a balanced diet
 - (ii). The 3 basic food groups and their functions.
3. **CONTROL:** Fade up signature tune10seconds...then fade out.
4. **RADIO-TEACHER:** Hello standard 7 pupils. In today's programme we are going to learn about the basic food groups that make up a balanced diet. You will be required to give answers to our

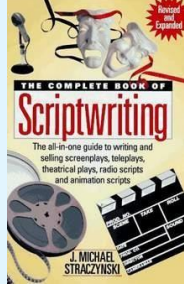
- questions to your class teacher when asked to do so. With us in the studio are radio pupils Irene, Catherine and Sarah.
5. **RADIO PUPILS:** Hello standard 7 pupils.
 6. **RADIO TEACHER:** We will start by asking you a question. What do you understand by the words 'balanced diet? [Repeat]. Tell your teacher now.
 7. **CONTROL:** Fade up bridge music ...20secs..... fade out
 8. **RADIO TEACHER:** Yes, Sarah.
 9. **SARAH:** A balanced diet is a diet which contains all the food nutrients required by the body, in our case carbohydrates, vitamins, proteins and fats.
 10. **RADIO TEACHER:** That is very good Sarah. Class, tell your teacher what you had for breakfast this morning. [Repeat]
 11. **CONTROL:** Fade up bridge music20secs.... fade out.
 12. **RADIO TEACHER:** Yes, Irene.
 13. **IRENE:** I had bread and butter, a fried egg, a cup of tea and an orange.
 14. **RADIO TEACHER:** Good! Irene had bread and butter, a fried egg, a cup of tea and an orange. She had foods which give energy, build and protect the body. Irene had a balanced diet.
 15. **RADIO TEACHER:** Children, now tell your class teacher examples of foods that give us energy. [Repeat]
 16. **CONTROL:** Fade up bridge music25secs..... fade out
 17. **RADIO TEACHER:** Now ...Catherine tell us some examples of energy giving foods? [Repeat]

18. **CATHERINE:** Energy giving foods are maize, cassava, potatoes and rice.
19. **RADIO TEACHER:** Good! Catherine has told us that examples of energy giving foods are maize, cassava, potatoes and rice. Others are fats and oils from butter, margarine, nuts and sugar. All these foods are carbohydrates. Now teacher....display the chart showing the various food groups. (Repeat).
20. **CONTROL:** Fade up bridge music.... 25secs..... fade out
21. **RADIO TEACHER:** Teacher, on your chart, show the pupils the group of foods that give us energy. [Repeat].
22. **CONTROL:** Fade up bridge music.....25secs..... fade out.
23. **RADIO TEACHER:** The teacher has just shown you foods that give us energy. The second group of foods are the body-building foods. Class, tell your teacher foods which build our bodies. [Repeat].
24. **CONTROL:** Fade up bridge music....25 secs..... fade out.
25. **RADIO TEACHER:** Yes, Sarah, tell us some of the foods that build our bodies.
26. **SARAH:** Body-building foods are milk, eggs, fish and meat.
27. **RADIO TEACHER:** Good! Some examples of foods the build our bodies are milk, eggs, fish, meat, beans, peas and groundnuts. These foods are also called protein-containing foods. They give proteins to our bodies. Now teacheron your chart show the pupils the group of foods the build our bodies. [Repeat].
28. **CONTROL:** Fade up bridge music.....25secs..... fade out
29. **RADIO TEACHER:** Your class teacher has shown you foods that build our bodies. Now look at the chart again. (Pause 7"). You can see the third and last group of foods. What name is given to this group of foods? Tell your teacher.....now.
30. **CONTROL:** Fade up bridge music....20 secs fade out
31. **RADIO TEACHER:** Yes, Irene.
32. **IRENE:** The third group of foods are called protective foods.
33. **RADIO TEACHER:** Protective foods provide our bodies with vitamins and mineral salts. Vitamins and mineral salts protect our bodies from certain types of diseases. Class.... Now tell your teacher some foods which protect our bodies.

34. **CONTROL:** Fade up bridge music....25 secs..... fade out
35. **RADIO TEACHER:** Yes, Catherine.
36. **CATHERINE:** Protective foods are carrots, tomatoes, vegetables, pawpaw and mangoes.
37. **RADIO TEACHER:** Protective foods protect our bodies from certain diseases and include carrots, tomatoes, vegetables, papaws and mangoes. Now, before we end our radio lesson let us again look at the main points of the lesson. We learnt that foods can be classified into three mains groups. What were these groups? Tell your teacher now.
38. **CONTROL:** Fade up bridge music....20 secs..... Fade out
39. **RADIO TEACHER:** Yes, Sarah.
40. **SARAH:** The three main groups are energy-giving foods, body-building foods and protective foods.
41. **RADIO TEACHER:** Good! The three main groups are carbohydrates or energy-giving foods.....proteins or body-building foods, and protective foods which give us vitamins a d mineral salt. These foods if eaten give us a balanced diet. Not, tell your teacher examples of energy giving foods. (Repeat).
42. **CONTROL:** Fade up bridge music..... 10 secs.....fade out
43. **RADIO TEACHER:** Yes, Irene.
44. **IRENE:** Energy-giving foods include maize, cassava, potatoes and rice.
45. **RADIO TEACHER:** Good! Now Tell your teacher examples of foods which build the body.
46. **CONTROL:** Fade up bridge music.....15 secs..... fade out
47. **RADIO TEACHER:** Yes, Catherine.
48. **CATHERINE:** Body-building foods include milk eggs, fish, meat, beans, peas and groundnuts.
49. **RADIO TEACHER:** Good! Class.... finally tell your teacher examples of protective or vitamin-giving foods. [Repeat].
50. **CONTROL:** Fade up bridge music.....15 secs..... fade out
51. **RADIO TEACHER:** Yes..... Sarah.
52. **SARAH:** Protective foods include carrots, tomatoes, vegetables, pawpaw and mangoes.
53. **RADIO TEACHER:** From what we have learnt, it is important to include the three groups of foods in our **daily** meals. It is only then that we can say we have a balanced diet. A balanced diet ensures we have enough energy, we build our bodies and protect our bodies from diseases.
54. **RADIO TEACHER:** In our next radio lesson, we shall learn about food preparation and serving. Between now and then your class teacher should arrange a visit to a nearby hotel to enable you to get an idea of how food is prepared and served. You can observe these at home. And that is all for today.
55. **CONTROL:** Bring up signature tune10 secs...then fade under.
56. **ANNOUNCER:** You have been listening to programme number 4 in the Primary Home Science series for standard 7. The title of the programme was “Basic food groups for a balanced diet.” This programme was written by Raphael Murruu, Selina Ominde, Eunice Obat and Gordon Oluoch. It was edited by Lordvicus W. Olanga and produced by Eunice Obat. Your presenters were Muruu, Ominde, Oluoch and Ogolla, and behind the controls was Jacob Oloo. The programme was produced in conjunction with the department of Educational Communications and technology, Kenyatta University.
- Until next week, it's bye from us.
57. **CONTROL:** Fade up signature tune20 secsthen fade out.

The sample script shown above could be improved by say inserting an interview with a nutritionist or school matron, or including a conversation between a housewife and one of the presenters. As a means of enabling the classroom teacher to utilize the audio material effectively, the Teachers' Notes on the topic should be availed.

E-tivity 9.2.3 Script writing

Numbering, pacing and sequencing	9.2.3
Title	Scriptwriting
Purpose	The purpose of this e-tivity is to enable you to prepare the audio script
Brief summary of overall task	Read the content in the notes especially the sample teachers' notes and the sample script.
Spark	
Individual task	After Reading material in the slide share and watching the video: a) From one of the reference books you intend to use in teaching the identified sub strand, select a paragraph and rewrite it in the radio language. b) Ensure the paragraph as rephrased sounds conversational, is of the right length and is free of unfamiliar terms c) Discuss • The need to edit the audio script to the expected duration using rehearsals • The need to choose an appropriate signature tune to the sub strand or subject • The need to choose an appropriate bridge music • The importance of repeating learner's answers □ The importance of repeating questions, definitions d) Prepare an audio script that would give a recording of 15 minutes (a rehearsal would be useful). e Ensure you incorporate learning activities in the programme f Save the work in your portfolio

Interaction begins	g) Using the sample script for comparison, check whether your script meets the expected standards. h) Post your work in the discussion forum i) Read what your colleagues have posted k) Read comments from your facilitator to get feedback l) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take two hours
Next	Recording and editing of the recording

9.2.4 Recording and Editing of the Recording

Once the script has been written, edited and all the inserts and sound effects lined up than the next stage is to record the content (script). The producer (that is the supervisor of the recording) must use voices that have been auditioned. During auditioning the voices are chosen for their quality in terms of pronunciation, clarity, fluency, pace, intonation and general voice quality. The voices should also be chosen for their suitability for narration, drama, poem reciting, and radio pupil roles.

Once the producer has assembled his artists (presenters) in the studio, each of them should have a copy of the script and thoroughly rehearse the script several times. This will help the producer and technician get an estimated duration of the programme. If there is need for further editing it can be done at this stage to reduce the programme to the prescribed duration and still have the message. During the

recording session, the technician should ensure that the microphones are properly positioned to maintain a good recording level throughout the session.

If the duration of the recorded programme with all the inserts, music bridges and pauses included is still higher than the prescribed time, then post-production editing can be done. This can be accomplished through mechanical splicing or dubbing.

Editing

Editing a material means removing parts of it and re-organizing the material to give it 'shape'. Editing can be done in two ways:

Script Editing

Before recording a programme the writer should allow a colleague to read through the script and give comments. A media specialist should also be allowed to judge the script for its suitability. The writer can then modify the script in line with the comments given by the colleague and the media specialist. Sometimes the time slot allocated for the programme is 20 minutes and on rehearsal the programme overshoots the time. It is then necessary to cut down on the quantity of material to bring it down to under 20 minutes.

Tape Editing

This is an important aspect of recording. It is conveniently done in tapes than in cassettes. This is because of two reasons. Firstly, the tape speeds

7 ½ and 15 inches per second (is) are ideal for editing than the slower audio cassette speed of 1 7/8 is. Secondly the tape is one track while the cassette is two track. The following alterations can be done on tape programmes through editing:

- (i). Reducing the overall length of the programme

- (ii). Re-arranging the order of events by placing one sound sequence ahead or behind its original position on the tape, thus providing bases for comparisons, contrasts or emphasis.

- (iii). Inserting new material such as speeches, music etc.

There are two methods of tape editing:

Electronic Editing (Dubbing)

This is ideal for audio cassettes and involves copying (dubbing) one recording through a second recorder onto another tape or cassette. This is done without cutting the original tape / cassette.

This method must be used very carefully as it introduces pops at the start and end of a sound sequence

Mechanical Editing (Splicing)

This method involves using scissors or tape slicer and splicing the tape. It is highly recommended for ‘polishing’ an interview by removing slight imperfections such as mmm eer, to add tape for lengthening pauses, reducing pauses or even to substitute a corrected bit of narration. The steps to follow are:

- (i). With the script in front of you listen to the recorded tape, listing spots that need editing and record the counter reading.
- (ii). Dub the tape onto another fresh tape, file the original tape for reference. Proceed with the tape you have just recorded.
- (iii). Lace the freshly recorded tape onto the recorder Play it and confirm the counter readings recorded earlier.
- (iv) Carry out rough editing first, that is removing whole paragraphs, whole sentences and rearranging them in the appropriate order. To identify a point to cut we engage the ‘edit’ key on the tape recorder, then we can move the spools manually back and forth across the play head. (v) The point to be cut can be carefully marked with a fine tipped felt pen or a china marking (grease) pencil.
- (vi) Two points on the tape can be connected using the splicing tape > the splicing tape should be put on the shiny side.
- (vii) After making a joint, playback the tape over the joint and satisfy yourself that it is well done.
- (viii). If a new bit is to be added, identify the spot where you wish to add the new bit. Mark it with a felt pen.
- (ix). Playback the tape over the two joints and satisfy yourself that the joints are well done and that the new bit fits in very well.

E-tivity 9.2.4 Recording and editing of the recording

Numbering, pacing and sequencing	9.2.4	
Title	Recording and editing of the recording	
Purpose	The purpose of this e-tivity is to enable you to record and edit the <u>recorded</u> audio programme	
Brief summary of overall task	Read the content in the notes and assemble the cast. Record the audio programme. Edit the audio programme to the expected duration.	
Spark		

Individual task	After Reading the notes: a) Suggest possible support materials for the programme. b) Check the audio script for any errors that may need correction c) Supervise the cast as they rehearse the presentation, keeping an eye on the duration d) Record the programme on a compact disc, memory card or flash disc e) Playback the recording and listen to it. e) Check the duration of the recording to ensure it is within the expected duration f) Hand in the recorded material, the script, and the Teachers' Notes g) Save the work in your portfolio
Interaction begins	h) Discuss the requirements of presenters in terms of voice quality, suitability of voice to the role and the need to pace the content appropriately for maximum learning i) Post your work in the discussion forum j) Read what your colleagues have posted k) Read comments from your facilitator to get feedback l) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This task should take two hours
Next	Operating audio visual equipment

9.2.5 Operating Audio-Visual Equipment

Since the classroom teacher can decide to use instructional materials at any time during the teaching it is necessary for him /her to know how to operate the large variety of equipment. The teacher should not be over dependent on a technician as the institution may not be endowed enough to afford the needed technical staff. Since the institution may have various equipment, they should have a copy of a manual of operating instructions for each model of equipment, the manual contains details of equipment features and operating techniques. It is important that all users maintain a high standard of performance, this leads to the realization of the following benefits:

1. Equipment can be kept to in working condition for a maximum amount of time
2. The cost of equipment maintenance, repair and replacement can be minimized
3. Incidents of the equipment failing to give the expected output can be reduced.

The sound produced should as much as possibly resemble the original sound. The quality of the sound produced is affected by:

1. The type and quality of equipment selected
2. Operator's skill in using the equipment.

Factors that affect the quality of sound reproduced during playback include:

1. Characteristics of the original sound
2. Environmental conditions in recording / playback
3. Capability of the recording medium to capture all the frequencies of the original sound.
4. Capability of the playback system to reproduce program material in full frequency and without distortion.
5. Ability of the operator to manipulate the equipment
6. Matching. All the components in an audio system should be of equal quality and have similar performance characteristics.

For operating most of the equipment, the teacher as well as the pupils must master the use of controls such as the normal play, record, fast forward (speed selection), pause, rewind and stop buttons. In some cases, the controls may be manual as well as through a remote-control device.

E-tivity 9.2.5 Operating audio visual equipment

Numbering, pacing and sequencing	9.2.5
Title	Operating audio visual equipment
Purpose	The purpose of this e-tivity is to enable you to acquire skills of operating commonly used audio visual equipment
Brief summary of overall task	Read the content in the notes
Spark	

Individual task	After Reading material in the notes: a) Distinguish between software and hardware in audio visual materials b) Discuss reasons why the teacher should have equipment operation skills c) Discuss the factors to consider when purchasing equipment d) Why should learners be involved in operating audio visual equipment e) Save the work in your portfolio
Interaction begins	f) Using short clear sentences, discuss the factors that must be taken into account for effective use of audio visual equipment. g) Post your work in the discussion forum h) Read what your colleagues have posted i) Keeping netiquette in mind, comment on at least two of their posts j) Read comments from your facilitator to get feedback k) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take two hours
Next	Locating sources of audio and audio visual materials

9.2.6 Locating Sources of Audio and Audio-Visual Materials

Having discussed most of the audio and audio visual resources, we have seen that some of them could be designed and produced by the teacher and / or pupils, it is advisable to note that they are also available at the following locations:

- 1 Kenya Institute of Curriculum Development (KICD)
- 2 The British Council Libraries
- 3 Teacher Advisory Centre
- 4 Kenya National Library Services
- 5 Learning Resource Centre of the local Teacher Training College.
- 6 Community Media Centre (if there are any)
- 7 Media Centre in other schools.
- 8 Community resources available within the locality such as factories.

E-tivity 9.2.6 Locating sources of audio and audio-visual materials

Numbering, pacing and sequencing	9.2.6
Title	Locating audio- and audio-visual materials
Purpose	The purpose of this e-tivity is to enable you to identify sources of audio- and audio-visual materials

Brief summary of overall task	Read the content in the notes and in other references.
Spark	
Individual task	After Reading material in the notes and referring to other readings: - Visit one of the locations where instructional materials can be found. - Enumerate the broad groups of audio-visual materials that can be found at the location. - Identify a source of resources within your locality - Save the work in your portfolio
Interaction begins	- Using short clear sentences, describe the challenges you would face when planning to teach one of your teaching subjects near the facility you had visited. - Post your work in the discussion forum - Read what your colleagues have posted - Comment on at least two of their posts - Read comments from your facilitator to get feedback - Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This task should take one hour
Next	Presentation

9.3 Assessment Questions

- In one of your teaching subjects, select a sub strand which in your opinion needs audio support.
- Prepare the teachers' notes
- Suggest possible support materials for the programme.

- (d) Prepare the audio script and check the audio script (editing) for any errors that may need correction
- (e) Supervise the cast as they rehearse the presentation, keeping an eye on the duration
- (f) Record the audio programme on a compact disc (CD), memory card or flash disc
- (g) Playback the recording and listen to it.
- (h) Check the duration of the recording to ensure it is within the expected duration
- (i) Submit the recorded material, the script, and the Teachers' Notes

9.4 E-Refereces

1. 3 Online Teaching Tools for Teachers (FREE!)
<https://www.youtube.com/watch?v=8C7KxoaU0h4&t=64s>
2. 7 Zoom Screen Share Tips Every User Should Know!
https://www.youtube.com/watch?v=25Awq_v1Zms
3. Audio Visual Aids –Meaning, Definition & Classification
<https://www.youtube.com/watch?v=MGobKltZtII>
4. Audio Visual Materials different types <https://www.youtube.com/watch?v=oO0LjWFz-0A>
5. Criteria for choosing audio visual aids
<https://www.youtube.com/watch?v=TeLdccfEPqY>
6. Google Classroom in 2020 The Complete Overview for Online Teaching
<https://www.youtube.com/watch?v=IeMq39WSuu8>
7. How To Make a Video Using Powerpoint (and get your FREE copy of my \$299 Coursepreneur™ program) https://www.youtube.com/watch?v=AGSk_ynrvTA
8. How To Make a Video Using Powerpoint 2010 Presentations (Tutorial For Beginners To Create a Video) <https://www.youtube.com/watch?v=fvkw-g0ilY4>
9. How to Make an Effective Audio-Visual Presentation
<https://www.youtube.com/watch?v=oFgoJqmylA>
10. How to present lectures on Google Meet - For Teachers
<https://www.youtube.com/watch?v=tdBTDrrj0Ro>
11. HOW TO RECORD VIDEO LECTURES | Using OBS for Online Teaching
<https://www.youtube.com/watch?v=zTjVBlnEiNI>
12. How to Record Yourself Presenting a Powerpoint Presentation
<https://www.youtube.com/watch?v=6abWf-5u3Gs>
13. How to Use Google Classroom for Remote Teaching
<https://www.youtube.com/watch?v=uUOwyIgzFOU>
14. HOW TO USE GOOGLE MEET | Video Conferencing Tutorial for Beginners (FREE for Everyone) <https://www.youtube.com/watch?v=IRZckVNQS7I>
15. How to use Zoom for Remote and Online learning
<https://www.youtube.com/watch?v=9guqRELB4dg&t=56s>
16. Teaching aids / characteristics of teaching aids / principles of teaching aids / demerits of teaching aids <https://www.youtube.com/watch?v=lWldaog5Ix8>
17. Teaching aids/audio visual aids/ importance of teaching aids/types of teaching aids.
<https://www.youtube.com/watch?v=lWldaog5Ix8>

18. THE best teaching tool for your classroom in years!
<https://www.youtube.com/watch?v=j6EwmrFhogA&t=47s>
19. Use of Audio - visual aids in Training
<https://www.youtube.com/watch?v=hWL6nkDck5o>
20. Visual Aids (B.Ed) https://www.youtube.com/watch?v=In0kFaWSg_Q
21. What is AUDIOVISUAL? What does AUDIOVISUAL mean? AUDIOVISUAL meaning, definition & explanation <https://www.youtube.com/watch?v=W08-ktWdyqA>

LESSON 10

COMMUNITY RESOURCES WITH SPECIAL EMPHASIS ON REALIA (Real Things)

10.1 Introduction

Firsthand experiences offer opportunity for effective learning. Through firsthand experiences, learning becomes more interesting, concrete and life-like. Firsthand experience also leads to the development of a wide range of important skills such as creativity, observation and many social relating skills. One way of involving learners in firsthand experiences is through the use of **community resources** and one important component of community resources is the use of **realia** or real things. This lesson will deal with how to plan for and effectively use real things in the teaching/learning process.

10.1.2 Learning Outcomes

By the end of this lesson you should be able to:

10.2.1 List the various categories of community resources.

10.2.2 Define the term realia.

10.2.3 Suggest various kinds of realia relevant to the learning of various topics in your subject area/s.

10.2.4 Outline the process from collection to presentation of realia in class.

10.2.5 Plan and use realia to teach a topic relevant to a particular class-level.

10.2.1 Type of Community Resources

Community resources are common-place experiences that are found within a community around a school. Community resources enable a learner get firsthand contact with the reality of issues to be learnt. Community resources can be categorized into four groups; namely, people, places, activities and things (realia). In this lesson we shall focus on the skills of preparing and presenting realia for purposes of teaching and learning.

10.2.2 Realia or Things or Real things

Realia can be variously referred to as things, objects, real things, specimens or simply things. The technical term, realia is usually used in educational technology to refer to objects, things specimens and other artifacts found within a community that can be used to illustrate, compare or clarify an idea or concept.

The ranges of things that can be used to aid the learning of various topics in various subject areas are enormous. Examples include: animals (dead or alive), animal parts, plants, antiques, tools, foods, household objects, art objects and so on.

E-tivity 10.2.2 Realia

Numbering	10.2.2
Title	Realia
Purpose	The purpose of this e-tivity is to enable you recognize the essence of community resources and realia in the teaching/learning process.
Summary of overall task	Watch this video , and then make a list of various kinds of realia you would suggest for the teaching of a topic in one of your teaching subjects.
Spark	
Individual contribution	(a) List as many real objects as you can that may be relevant to the effective teaching of the named topic in your subject area.
Dialogue begins	(b) Receive responses from individuals.
E-moderator interventions	• Critique the individual contributions. • Reinforce correct responses, summarize and close the e-tivity
Schedule and time	Total study time will be 1 hour.
Next	The next activity will involve planning a lesson that makes use of realia.

10.2.3 Collecting, Preparing and Presenting Realia for Teaching

It is necessary for you to be familiar with the kinds of items within the community around your school that can be used to teach different topics in your subject area/s. Many of the items that could be used as realia are around us as part of nature, household goods, or even hobby items. When possible, it is advisable to involve your learners in the collection of the items. Collection of the items should be done in a systematic manner, involving:

- (a) Planning
- (b) Research
- (c) Preservation
- (d) Mounting
- (e) Labeling and

(f) Displaying in interesting ways.

10.3 Assignment

1. Prepare realia relevant to a topic in one of your teaching subjects and apply as much of the following processes as possible:

- (a) Research
- (b) Preservation
- (c) Mounting
- (d) Labeling
- (e) Displaying in an interesting manner

10.4 E-References

Armstrong, Ginger. Realia Teaching. Retrieved from https://www.youtube.com/results?search_query=realia+teaching+strategy%3A+ginger+armstrong&pbjreload=101

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International TEFL Academy, (16th Dec.2016). Retrieved from <https://www.youtube.com/watch?v=SHIp5bRemso&t=25s>

Kristie and Jess. Bug Box Tutorial. Retrieved from <https://www.youtube.com/watch?v=YhLZKTUeQpc>

Ruhl, Joe. Teaching Methods to Inspire the Students of the Future. Retrieved from https://www.youtube.com/results?search_query=teaching+methods+for+inspiring+the+students+of+the+future+ +joe+ruhl

LESSON 11

CONSTRUCTION SKILLS

11.1 Introduction

In this lesson, you are going to deal with construction skills. Construction skills apply to the production of models to be used in to instruct learning. The relationship between models and real things in that, models are recognizable three-dimensional (3-D) likeness or representations of real things or objects. You are already familiar with the skills of preparation of real things (realia) which you practiced in the previous lesson.

11.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

11.2.1 Explain the characteristics of 3-D media.

11.2.2 Explain specific circumstances, in your subject area, that may warrant the use of models as opposed to graphic materials and realia (real things).

11.2.3 Suggest possible uses of special models; namely, dioramas and mockups to teach specific content in your subject area.

11.2.4 Use locally available materials to produce a model for teaching particular content relevant to your teaching subject.

11.2.1 Characteristics of Models

1. They are Three Dimensional (3-D) in that they possess height, length and thickness. This is unlike graphic material that are Two Dimensional (2-D) and hence only poses height and length and lack thickness.
2. They are usually enlargements or reductions of objects to an observable size.
3. They may also provide an interior view of objects. For example, a model of a tooth will reveal the layers of enamel and dentine and nerves which would otherwise be difficult to visualize or see.
4. They simplify complex objects and help in explaining operations; such as, the workings of an internal combustion engine. How a motor engine works.
5. They accent or emphasize important features by adding colour or texture to a model. This enhances the effective identification of important features. Conventionally, for example, for purposes of identification, arteries are usually coloured bright red and veins bright blue.

6. Some models are usually made in such a way that they can be assembled or disassembled to reveal its inner and outer features. For example, the heart and how it functions.

11.2.2 Circumstances Requiring the Use of Models

As rule of thumb, the priority is always to first and foremost use real things for you to illustrate your lesson whenever applicable. However, there are certain circumstances when our only recourse is to use models or a combination of real things and models. Some of the circumstances are as follows, when real things are:

1. Difficult to come by
2. Too large to realistically use in a classroom, for example the solar system.
3. Too small to meaningfully present in class, for example an atom or molecule.
4. Pose safety problems.
5. Does not provide effective illustration of its working or operations, for example the process of breathing and synchronization of lungs, diaphragm and ribcage.

11.2.3 Special Types of Models

Mock-ups

A mock-up is an operating model usually at full scale, designed to be worked with directly by learners for specific training or analysis.

Mock-ups have many applications in training including devices for training drivers and aero plane pilots. Another example is baby-sized dolls to train in say how to correctly bath a baby.

In the cases above you will notice that mock-ups help in stimulating the actual situation and that by its nature it can be used without risking the life of the trainee as well as others.

Dioramas

A diorama is a three-dimensional (3-D) scene in depth incorporating a group of modeled objects and figures in a natural setting.

A diorama is usually set up in a miniature stage with a group of modeled objects and figures in the foreground which is blended into a realistically

painted background. You can use it to teach about say the setup of a traditional African homestead in social sciences, an ideal small-scale farm in Agriculture or to show a proposed architectural project. These examples of dioramas can also lend themselves to teaching in subject areas for which they were not initially intended. For instance, the traditional African homestead or an ideal farm can be used in language learning to teach of vocabulary, grammar, and so on.

E-tivity 11.2.3 Using 3-D Models for Instruction

	11.2.3
Title	Using 3-D Models for Instruction
purpose	The purpose of this activity is to improve your decision-making skills in sensing possibilities as regards models that are likely to lend themselves favorably to learning some content in your subject area.
Summary of overall task	Watch this video by BioMedical Art.
spark	
Individual task	(a) Suggest why a named model would be a better aid, rather than realia or a graphic material, to teach some specified content in one of your teaching subjects. (b) What advice would you give to a science teacher who is unwilling to use a model to teach the workings of a heart?
Interaction begins	(c) List the various reasons and circumstances that may necessitate the use of models over realia and graphic materials. (d) Using the heart as a specific model, list the various reasons why it would be a better teaching aid than a real heart or a drawing/diagram.
E-moderator interventions	• For each task, encourage and list as many reasons as possible from the floor; thereafter. • Revisit each of the suggestion above and evaluate their merits and demerits. • Close the activity

Schedule and time	This activity should take one hour.
Next	Production of a model to teach some content in the student teachers' subject area. The model should be well made. The model should be handed-in accompanied by a brief description of what for, how and when it will be used in the course of a lesson.

11.2.4 Some Production Considerations

1. Models can be made using a wide array of materials most of which are often discarded. As such, you will spend little or no money and there is no reason why you should not make a model whenever it is required.
2. Examples of materials for modeling include: Paper (stiff, plain, used paper packages and boxes including cartons); pieces of cutout or left-over clothes, wood scraps, wire (coat hangers), rapier Mache, Styrofoam, clay, plastic (bottles, cups, plates), hardboards, cardboards, and so on.
3. In addition to discarded materials, you may find it necessary to buy some basic materials such as glues (paper glue, white wood glue, shoemakers' glue, and so on); adhesive tapes (cello-tape, masking tape, insulating tape and so on); paints of different kinds depending on the required work (gloss or oil-based paints, water-based paints, colored pieces of paper or cloth to introduce color); vanishes; turpentine to thin oil-based paints; and many more.
4. You will also need a variety of tools some of which include, scissors, staplers and staples; carpentry tools (hammers, chisels, saws, and so on); metal work tools (pliers, rivets, drills, and so on); tailoring and leatherwork tools; and many more.
5. Should you, as a teacher feel inadequate in skill to produce a model that you have conceived, then, by all means engage a competent craftsman. It is essential that the quality of models produced are of as high quality as is possible. Note that the production quality of a teaching aid has an effect on the quality of learning it will elicit. Poor teaching aids, even when appropriate to what is being learned, often results in ineffective learning.
6. In addition to models being well made, they should be attractive to look at, durable and suit the intended purpose.
7. Make use of well selected colors and color combinations. Color attracts attention and highlights important details.
8. Use your creativity to come up with models that can be used to teach content in your subject area.
9. Since there is a cost factor (in terms of time, money and effort) in the production of some of the models, it is necessary that the models are stored well so that they can be re-used in years to come. The models and any other

resource that you make should be catalogued and stored in a school learning resource centre (LRC). Refer to the next chapter.

11.3 Assignment

Make a model to teach some content in one of your subject area. The model should be well made. Handed in the model together with a brief description of what for, how and when it will be used in the course of a lesson.

11.4 References

BioMedical Art, https://www.youtube.com/watch?v=vhl0n_pStYg&t=281s

LESSON 12

MANAGEMENT OF THE LEARNING RESOURCE CENTRES

12. 1 Introduction

Every institution needs space designated to design, produce and avail for use the teaching/ learning resources. This unit looks at the learning resource centre (LRC) concept. It emphasizes resource-based learning. It stresses human and non-human resources. The unit also considers the role of external as well as internal publics in the management of learning resources

12.2 Lesson Learning Outcomes

By the time you get to the end of this unit you will be able to:

12.2.1 Define a Learning Resource Centre

12.2.2 Give the rationale of setting up a Learning Resource Centre

12.2.3 Discuss the functions of a Learning Resource Centre

12.2.4 Spell out the services provided by the LRC

12.2.5 Discuss the various inputs necessary in the proper management of the LRC

12.2.6 Discuss some of the problems facing the LRC

12.2.7 Suggest helpful guidelines that can be implemented to improve management of the LRC

12.2.1 Definition of a Learning Resource Centre

A Learning Resource centre is invariably referred to as Educational Media Centre, Multi Media Centre, Instructional Materials Centre, or Teachers Advisory Centre. It is a well-planned area housing materials and equipment for instructional use under the supervision of a competent person. This person is the Media resource specialist usually well versed in general education and educational materials.

A Learning Resource Centre can also be said to be a store house of materials from which people can learn what they want to learn with a fair chance of doing it well. The purpose of the LRC is to improve instruction through service to teachers and students.

A Multi media centre is a place for stimulating the active creation and use of resources including production of homemade resources, selection / acquisition of

resources, classification and indexing for easy retrieval, storage use, evaluation and weeding.

12.2.1 E-tivity

Numbering, pacing and sequencing	12.2.1
Title	Definition of LRC
Purpose	The purpose of this e-tivity is to enable you to distinguish between LRC, multi media centre, library and a Teachers advisory centre. You will also be able to state the purpose of the LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube
Spark	
Individual task	After Reading material in the notes and related references: a) State in your own words what you understand by the LRC b) Save the work in your portfolio
Interaction begins	c) Using short clear sentences, describe three characteristics of developmental tasks. d) Post your work in the discussion forum e) Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take one hour

Next	Rationale for setting up the LRC
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12.2.2 Rationale for Setting up a LRC

A Learning Resource Centre is generated from the recognition of three facts: □ Learning takes place through a variety of channels or avenues. Some ideas cannot be reliably communicated through books. Music for example must be heard, paintings must be seen perfumes must be smelt etc. This implies that any time two or more media are used to teach, multimedia instruction occurs.

- Individual differences. People are different from one another. Some learn things best by reading, others by hearing, and others by writing and most by combining two or more activities.
- Need for private learning. Whilst some people learn best in groups, others on their own, and generally speaking we all need some kind of private learning from time to time.

E-tivity 12.2.2 Rationale for setting up a LRC

Numbering, pacing and sequencing	12.2.2
Title	Rationale for setting up a LRC
Purpose	The purpose of this e-tivity is to enable you to justify the setting up of a LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources
Spark	
Individual task	After Reading material in the notes and related references: - Prepare a report justifying the setting up of the learning resource center in an institution. - Save the work in your portfolio

Interaction begins	c) Using short clear sentences, discuss three reasons why an institution needs a LRC. d) Post your work in the discussion forum e) Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take one hour
Next	Functions of a LRC

12.2.3 Functions of a Learning Resource Centre

A Learning Resource Centre serves the following functions:

- As depository for special materials which aid the teacher to present better their subject areas and to take care of the wide range of individual differences that exist in classes or schools or colleges of teacher education. The educational aids adaptable to further the professional growth of the teaching staff form part of this organization. The LRC serves as a central source of instructional materials, providing economy of teacher time and effort and encouraging simultaneous use of all types of materials.
- As a place where new or experimental types of instructional materials may be developed, produced and evaluated for possible future use or adoption. It is a place for implementing newer media in education through ready access to materials.
- As a place where circulation of audio-visual materials, coordination of a variety materials and in-service education of teachers is carried out. It has professional consultants who are experienced in the selection of all types of materials.
- Conducting in service education for serving teachers.

Broadly speaking, a fully-fledged LRC has three roles namely:

- The storage and retrieval of learning media
- Creation and production of learning materials
- Instruction and advice on the utilization of the facilities available.

Specifically, the LRC should be organized to provide:

- Catalogues and inventories of all types of resources
- Maintenance and service of all the teaching tools

- Information to teachers and students about new developments in materials, equipment and teaching technology
- Produce materials which are unique to a specific teaching / learning situation
- Provide assistance in the locating of needed teaching / learning materials
- Assist teachers and students in the use of teaching equipment and materials
- Provide space and facilities for teachers and students to pre view, audition, review and try out various teaching / learning media
- Serve as a comprehensive learning laboratory in which students can learn to use all types of learning materials and equipment
- Provide for continuous evaluation of the programmes and services.

12.2.3 E-tivity

Numbering, pacing and sequencing	12.2.3
Title	Functions of a LRC
Purpose	The purpose of this e-tivity is to enable you to describe the functions of a LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources
Spark	
Individual task	After Reading material in the notes and related references a).In your own opinion enumerate the functions of a LRC. b) Save the work in your portfolio

Interaction begins	c) Using short clear sentences, discuss three reasons why an institution needs a LRC. d) Post your work in the discussion forum e) Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take one hour
Next	Organization and management of the LRC

12.2.4 Organization and management of the LRC

The organization and management of the LRC is best discussed under three areas namely personnel, physical or space provision and material requirements.

Personnel in the LRC

This is built on a kind of three-dimensional plan with the Director of the centre as the warden. S/he is a specialist in media education. Also, there should be support staff comprising technical staff, graphics and secretarial staff, librarians and teachers. Their functions should be to provide students with materials from which they can learn on their own to set up and if necessary, operate equipment to produce learning materials, to provide support to teachers by either retrieving or preparing materials or by giving assistance and advice in the actual presentation of material. They are also involved in curriculum development.

Staff training

New staff should be in-serviced in order to take full advantage of the facilities which are provided. Every member of staff should be trained by the media specialist to operate the electronic equipment and any other equipment s/he may not be familiar with.

E-tivity 12.2.4 Personnel in the LRC

Numbering, pacing and sequencing	12.2.4
Title	Personnel in the LRC
Purpose	The purpose of this e-tivity is to enable you to identify and assess quality of personnel required in a LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources
Spark	
Individual task	After Reading material in the notes and related references. a) Develop a checklist and use it to assess the suitability and availability of the personnel in the learning resource center within your neighborhood. b). Save the work in your portfolio
Interaction begins	c). Using short clear sentences, discuss the quality of personnel in a LRC you are familiar with. d). Post your work in the discussion forum e). Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take one hour
Next	Physical of space provision in a LRC

12.2.5 Physical or space provision in the LRC

When setting up an LRC consideration should be given to the provision of spaces for:

1. Periodicals / Journals room in which current copies are on display and where back numbers issues of journals can be borrowed.
2. Learning resources room containing carrels at which students can see slides, listen to tapes and CDs, study video / VCD materials without

interfering with other users. These can also be transmitted on request from a control room.

3. Control room in which sound tapes/CD, records and video/VCD can be played and transmitted to terminal outlets in most rooms within the institution.
4. Recording room for the making of sound recordings and the dubbing of films.
5. Television studio which although can be well equipped is a classroom converted for this purpose. Mobile television trolleys which can be operated by a member of staff working on his/her own with a class in any room.
6. Media workshop in which students and staff can produce their own visual aids and other teaching / learning aids.
7. Reading area cum space for seminars or group discussions usually designed to fit 10 people.
8. Maintenance facilities. The LRC should have space and personnel available to carry on the maintenance programme.
9. Graphics artist studio in which a graphics assistant produces visual aids of professional quality.
10. Reprographics room where printed or duplicated teaching materials and handouts are prepared.
11. Conference room large enough to accommodate groups of various sizes.
12. Cybercafé where staff and students can surf the internet and access other related materials.

E-tivity 12.2.5 Physical or space provision in the LRC

Numbering, pacing and sequencing	12.2.5
Title	Physical or space provision in the LRC
Purpose	The purpose of this e-tivity is to enable you to assess the suitability of space in a LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources

Spark	
Individual task	After Reading material in the notes and related references a). In your own assessment, comment on the suitability and availability of space in a LRC. b). Save the work in your portfolio
Interaction begins	c) Using short clear sentences, discuss the quality of personnel in a LRC you are familiar with. d) Post your work in the discussion forum e) Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This task should take one hour
Next	Materials requirements in a LRC

12.2.6 Materials requirements in a LRC

Materials and equipment should be catalogued and entered into comprehensively prepared inventories. Some of these items include:

1. Curriculum materials and professional literature for teachers
2. Bulletin boards, chalkboards, display centres, a card index and an evaluation index
3. Pre view and laboratory facilities provision
4. Equipment for conditioning, rewinding, inspecting and splicing films. In addition, there should be provision for simpler repair of audio-visual equipment and materials.
5. Materials and equipment provided for teachers and students to make maps, charts, graphs, models, mock ups, objects, specimens, posters, puppets, dioramas, flannel boards, exhibits, slides and items needed for demonstrations and experiments

6. Tape transcripts for use in the classroom

E-tivity 12.2.6 Material requirements in the LRC

Numbering, pacing and sequencing	12.2.6
Title	Material requirements in the LRC
Purpose	The purpose of this e-tivity is to enable you to assess the variety of resources in a LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources
Spark	
Individual task	After Reading material in the notes and related references - Visit a learning resource center (or a library in a school in the neighborhood) and make a report on the variety of resources available. - In your own assessment, comment on the suitability and availability of materials in the LRC. - Save the work in your portfolio
Interaction begins	- Using short clear sentences, discuss the quality of personnel in a LRC you are familiar with. - Post your work in the discussion forum - Read what your colleagues have posted - Comment on at least two of their posts - Read comments from your facilitator to get feedback - Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion

Schedule and time	This task should take one hour
Next	Problems facing the LRC

12.2.7 Problems facing the LRC

Various problems face the LRC and hinder its use for effective use in enriching instruction. Some of these include:

- Availability of personnel of the right caliber. Currently there are very few media specialists and support staff and appointments leave a lot to be desired.
☐ Limited funds prohibit establishment of an LRC in schools and zones.
- Many schools do not have adequate space to house an LRC's sophisticated activities ☐ Time factor. The school day is only as long as is the school year.
- Attitude of the teaching staff has tended to be negative. Teachers tend to fear anything technological being introduced in teaching.

E-tivity 1.2.7 Problems facing the LRC

Numbering, pacing and sequencing	12.2.7
Title	Problems facing the LRC
Purpose	The purpose of this e-tivity is to enable you to find out problems facing the LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources
Spark	
Individual task	After Reading material in the notes and related references a). Interview personnel at the learning resource centre visited earlier and find out the problems faced by the institution in managing the facility. b). Save the work in your portfolio

Interaction begins	c) Using short clear sentences, discuss the quality of personnel in a LRC you are familiar with. d) your work in the discussion forum e) Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	- Ensuring learners are focused on the contents of the discussion - Encouraging learners (quiet ones) to contribute - Providing feedback/ teaching points - Closing the discussion
Schedule and time	This task should take one hour
Next	Helpful guides in the effective management of the LRC

12.2.8 Helpful guides in the effective management of the LRC

To manage the LRC effectively, it is important for the school to:

1. Know the cost of various items in the LRC
2. Know that the most expensive is not necessarily the best
3. Request for quotations frequently from different manufacturers
4. Make a complete survey of your resources to determine their present status
5. Ascertain need for new equipment and services based on demand
6. Have a list of priority items in case you may acquire funds (windfalls) for new purchases
7. Involve other teachers in decision making regarding planning, purchase and utilization of media and resources
8. There should be delegation of responsibility
9. Allot funds according to needs

12.2.8 E-tivity

Numbering, pacing and sequencing	12.2.8
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Title	Helpful guides in the effective management of the LRC
Purpose	The purpose of this e-tivity is to enable you to identify helpful guides needed for the effective management of the LRC
Brief summary of overall task	Read the content in the notes and search for related information from the YouTube and any other sources
Spark	
Individual task	After Reading material in the notes and related references a). After visiting a LRC and interviewing personnel at the learning resource centre, give suggestions on the way forward for effective management of the LRC. b). Save the work in your portfolio
Interaction begins	c) Discuss the way forward in improving management of the LRC. d) Post your work in the discussion forum e) Read what your colleagues have posted f) Comment on at least two of their posts g) Read comments from your facilitator to get feedback h) Identify and add the new ideas you have learnt to your notes/portfolio
E-moderator interventions	• Ensuring learners are focused on the contents of the discussion • Encouraging learners (quiet ones) to contribute • Providing feedback/ teaching points • Closing the discussion
Schedule and time	This task should take one hour
Next	Practice

12.3 Assessment Question

A local Community based organization (CBO) co-opts you into a committee to develop a proposal for establishing a Community Learning Resource Centre. Prepare a ten page proposal to be submitted to the CBO.

12.4 References

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ANSWERS TO LESSON ASSESSMENT QUESTIONS

1.5 LESSON ONE (INTRODUCTION TO THE UNIT, MEDIA PRACTICALS & PEER TEACHING)

Prepare a 10-minute micro lesson for the skill of lecture/explanation. Submit the lesson plan to your lecturer for assessment and feedback.

ANSWER

The micro lesson plan should be in the standard KU format i.e should have

- Administrative details such as – student teacher’s name, registration number, name of school, class/form, subject, topic , sub-topic, time **(2 marks)**
- Learning outcomes stated in measurable/ observable terms and generally satisfying the “SMART” acronym **(3 marks)**
- Column for time indicating the three stages namely introduction , lesson development and conclusion with time appropriately allocated **(2 marks)**

2.3 MARKING SCHEME: LECTURE SKILL

- (a) Lesson Plan:
- (i) Are objectives stated in:
Behavioral TERMS?
Are they SMART?

LESSON FORMAT: Has the learner indicated the four columns of the lesson, i.e. Time, Content, Learning Activity and Resources?

Time- is the time allocated to each part of the lesson, i.e. introduction, Development and Conclusion enough for the content indicated?

INTRODUCTION:

During this stage, has the student-teacher indicated whether he is using formal or informal skill?

Has the teacher introduce his topic/lesson in an interesting and captivating manner

DEVELOPMENT:

During this part, has the teacher put the content that shows that he will be the one talking and explaining the content he has prepared without learner’s interruption by way of asking and/or answering questions?

CONCLUSION:

At this stage, has the student teacher manifested that he would be the only one summarizing the lesson while the learners listen to him.

LEARNING ACTIVITIES:

Does this column show that learners are passive receivers and only participate in writing their own notes during the lecture?

RESOURCE MATERIALS:

Under this column, are there indications that the resources

Would be used only by the teacher, thus ensuring that the lesson is teacher-centered?

HOW TO IMPROVE LECTURE SKILL

- (I) Make it a formal lecture where you demonstrate, explain principles and concepts.
- (II) Prepare, organize and rehearse the lesson before presentation.
- (III) Employ other skills such as set-induction and stimulus variation as well as integrated skill.
- (IV) Ensure that you are eloquent and loud enough.
- (V) Dress modestly in order not to attract attention in a negative way.
- (VI) Change your voice when stressing a certain concept or point.
- (VII) Move about and avoid standing at one place for long.
- (VIII) Use eye-catching materials as teaching aids if time allows.
- (IX) Ensure the content is suitable to the level of the learners.
- (X) Allow learners to interrupt at intervals if they don't understand something.

3.5 LESSON THREE (QUESTIONING REINFORCEMENT)

1. LESSON PLANNING

- (a) The lesson plan should provide for active engagement of learners particularly through question-answer.
- (b) The content of the lesson should lend itself to questioning.
- (c) In the learning activities, the questions can be stated directly; or alternatively, it can be indicated what is expected of the learner. To explain, describe, suggest, rephrase, and so on.
- (d) Questioning should occur at all the stages of the lesson; introduction, body and conclusion.

2. QUALITY OF QUESTIONS

Check the questions for: (a) statement clarity, (b) good mix of questions ranging from lower-order to higher-order questions, (c) Relevance to content under review, (d) thought-provoking.

3. RESPONSES TO QUESTIONS:

- (a) probing/prompting, (b) re-directing, (c) Refocusing, (d) Reinforcing verbally, nonverbally or both.

4.3 LESSON FOUR (SET INDUCTION & STIMULUS VARIATION)

1. **Lesson Plan:** to indicate use of appropriate set to induct learners at the start of the lesson and at the start of a new activity;
The lesson to capture and sustain interest of learners in terms of among other aspects logical sequencing of content; teacher mastery of content, active engagement of learners; lesson takes into account learners' interests; Learning involves most if not all the senses, Interaction does not just remain teacher-learner, but also involves learner-learner.
2. **Sustenance of interest:** as given in Q.1 above plus any other/s

5.3 LESSON FIVE (SMALL GROUP WORK & USE OF EXAMPLES)

LESSON PLAN TO SHOW SOME OF THE small groupings discussed in the lesson and to also indicate use of examples to illustrate the lesson. Examples can be real things, models, specimens or ideas, etc

6.3 LESSON SIX (INTERGRATED SKILLS ASSIGNMENT)

- (a) Are there more than one objective?
- (b) Are the objectives stated in learners' behavior?
- (c) Are the objectives SMART?

APPROPRIATE COLUMNS:

TIME: Is the time given enough for the teacher's and the learners' performance?

CONTENT: Has the student-teacher indicated the three parts of the lesson, that is Introduction, Body/Development and Conclusion, and are they written or typed in upper case and underscored?

INTRODUCTION: Has the student-teacher shown various approaches he would use to introduce his lesson, for example greeting the learners, asking them to remind him what the last lesson was about or him revising the previous lesson by use of questioning the learners, etc.?

BODY/DEVELOPMENT: Are there any indication that the student-teacher would use Different approaches such as grouping students to discuss or Perform an activity, asking learners to perform an activity, Teacher explaining or using some visual aids during the Lesson, demonstrating, etc.

CONCLUSION: Has the teacher indicated how he would conclude the lesson
Using different approaches, e.g. asking and answering learner's
Questions, reviewing what has taught, giving an appropriate
Assignment, etc.

LEARNING ACTIVITIES: Is there indication that learners are active during the three
Parts of the lesson? During Development, has the student-
Teacher show how learners will be involved for example,
In writing some notes, asking and answering questions,
Discussing in groups, demonstrating what they have learnt, etc.

RESOURCE MATERIALS: Has the student-teacher shown the teaching aids he would
Use at each part of the lesson from Introduction to Conclusion
for example, real objects, maps, charts, text books, chalkboard
and any other realistic visual aid, including audio and videos.

7.3 LESSON SEVEN (WRITING & INTERACTIVE BOARDS)

1. Award 2 marks each for any three (3) writing boards discussed in the lesson.
(Max 6 MARKS)
2. Accept any good explanation on how to use the 3 boards mentioned in part 1. Above
(MAX. 4. Marks)

8.3 LESSON EIGHT (INSTRUCTIONAL GRAPHIC MATERIALS)

A	Production Quality	MAX Mark
	• Layout (effective use of working surface) • Lettering (Proportional, well-spaced and consistent in style) • Illustrations (Clear and suited for the intended purpose)	6 Marks
B	Practicability of the Graphic Material	
	Work is: • Bold and will be clearly visible in class • Not loaded with unnecessary details • Interpretable by intended audience	5 Marks
C	Adequacy of Usage of Materials as Evident in the Lesson Plan	

	Lesson plan logically shows: What for? When? and How? the graphic material will be used in the course of the lesson.	4 Marks
D	Originality	
	The Graphic material: • Explores other topics or information not often presented by peers • Uses unconventional production materials such as cloth, paints, and so on.	2 Marks
E	Overall Appearance and Care in Handling of the finished work	
	Work is: • Attractive enough to capture attention • Well preserved, tidy; not folded and creased	3 Marks
TOTAL		20 Marks

9.3 LESSON NINE (VISUAL AND AUDIO-VISUAL MEDIA)

- i. Teachers' notes with elements : Title , learning outcomes, summarized content, learning activities (before, during and after listening), and support materials
(2 marks)
- ii. The audio script
 - Layout
 - Inclusion of sound effects
 - Appropriate length to give between 15 and 20 minutes of recording
 - Inclusion of opening and closing announcement using signature tune
 - Use of bridge music as appropriate (2 marks for each item = 10 marks)
- iii. The recording
 - Sound quality and sound level
 - Voice quality
 - Pacing of content
 - Listener involvement (2 marks for each item= 8 marks)

10.5 LESSON TEN (PREPARATION OF REALIA)

Award MAX 5 marks for any of 4 item listed below that is presented in the model. MAX total 20 Marks

- (a) Research

- (b) Preservation
- (c) Mounting
- (d) Labeling
- (e) Displaying in an interesting manner

11.3 LESSON ELEVEN (3-D MODEL MAKING ASSIGNMENT)

A	Production Quality	MAX. Mark
	• Well made in that it can withstand handling over time • Makes use of well selected colors • Invites interest (appealing/interesting) • The quality of production is close to commercially made	5
B	Practicality	
	• Large enough to be seen by the whole class • Can be handled by learners • Helps learners understand the concept being taught	5
C	Creativity/Originality	
	• Uses ingenious/unusual material; such as colored paper, cloth, ect. • The model can be operated (a working model)	5
D	Adequacy of the model as evident in the accompanying write-up	
	The write-up explains clearly what the model will be used for, how and when it will be used in the course of the lesson.	5
E	TOTAL MARK	20

12.3 LESSON TWELVE (MANAGEMENT OF THE LEARNING RESOURCE CENTRE)

Proposal to have information on :

- i. Name of the CBO (1 mark)

ii.	Approximate population of catchment area mark)	(1
iii.	Needs assessment carried out marks)	(3
iv.	Needs of the population marks)	(2
v.	Source of funds for construction marks)	(2
vi.	Floor plan of the building showing spaces marks)	(3
vii.	Source of resources including categories of the resources. marks)	(3
viii.	Personnel requirements and their recruitment marks)	(3
ix.	Future funding of Resource centre and sustainability marks)	(2

Total 20 marks